

FIBONACCI–PELL NEAREST GAPS: AN ALL-EXPONENT CLASSIFICATION AND QUADRATIC-UNIT ORBIT RIGIDITY

DENYS DUTYKH AND LAURENT VUILLON

ABSTRACT. Let $\lambda = 1 + \sqrt{2}$, let (F_q) be the Fibonacci sequence, and put $D_{q,\sigma} = \sigma F_q + F_{q+1}\sqrt{2}$ for $\sigma \in \{1, -1\}$. For every $q \geq 1$ and every positive exponent n , we classify the strict factor-two gaps $\Delta = \lambda^n - D_{q,\sigma}^2$ whose quadratic norm equals minus the square of their coefficient content g . Exactly two occur: $(q, \sigma, n, g, \Delta/g) = (2, -1, 2, 6, \lambda^{-1})$ and $(4, 1, 6, 40, \lambda)$. In particular, no solution has odd exponent. By contrast, the underlying even factor-two windows occur with explicit positive densities governed by an irrational circle rotation.

The proof is exact and computer-assisted. In the even branch, a primitive factor rectangle supplies height separation; two logarithmic forms give $h = O(\log q)$ and $q < 10^{32}$, certified rational-interval reductions give $h < 192$ and $q < 90$, and exact recurrence arithmetic completes the enumeration. The odd branch has no such height separation, so a fixed-form-first bootstrap is required; it reduces to $q < 96$, where the terminal list is empty.

The map $S(u, v) = (v, u + 2v)$ is integrally conjugate to multiplication by λ through $A_{(u,v)} = (v - u) + u\sqrt{2}$. This gives an exact orbit/gap equivalence and a complete all-anchor classification for the two Fibonacci cores $(F_{q+1}, F_{q+1} + \sigma F_q)$: one canonical persistent family and three isolated time-zero hits are the only Pell-orbit intersections. Finally, a local rank criterion and Haar equidistribution produce simultaneous prescribed powers of compatible split primes in a natural plus-sign cross-gcd on genuine nearest windows, while the split prime 241 is obstructed. The work is motivated by the Markoff–Frobenius uniqueness conjecture, but no reduction from a hypothetical Markoff collision to these orbit families is claimed.

1. INTRODUCTION

How often can the square of a Fibonacci quadratic integer lie just below a power of the fundamental Pell unit, with the small difference pointing in a primitive negative-norm unit direction? This question combines an Archimedean nearest-neighbour condition with an exact norm–content condition. The former occurs with positive density; the latter will leave only two rows.

Let

$$\lambda = 1 + \sqrt{2}, \quad D_{q,\sigma} = \sigma F_q + F_{q+1}\sqrt{2}, \quad q \geq 1, \quad \sigma \in \{1, -1\}. \quad (1)$$

Suppose that a positive integer n lies in the strict factor-two window

$$D_{q,\sigma}^2 < \lambda^n < 2D_{q,\sigma}^2. \quad (2)$$

There is at most one such n , because the ratio of two distinct integral powers of λ is at least $\lambda > 2$. Write

$$\Delta_{q,\sigma} = \lambda^n - D_{q,\sigma}^2, \quad g_{q,\sigma} = \text{cont}(\Delta_{q,\sigma}), \quad (3)$$

where

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$$N(a + b\sqrt{2}) = a^2 - 2b^2$$

is the quadratic norm and the coefficient content of $a + b\sqrt{2} \in \mathbb{Z}[\sqrt{2}]$ is $\gcd(|a|, |b|)$. The target condition is

$$N(\Delta_{q,\sigma}) = -g_{q,\sigma}^2. \quad (4)$$

After division by its largest rational coefficient factor, the gap is then a norm- -1 unit. The two surviving examples already display the two possible orientations:

$$\lambda^2 - (-F_2 + F_3\sqrt{2})^2 = -6 + 6\sqrt{2} = 6\lambda^{-1}, \quad (5)$$

$$\lambda^6 - (F_4 + F_5\sqrt{2})^2 = 40 + 40\sqrt{2} = 40\lambda. \quad (6)$$

Our first main theorem says that there are no others, even when the parity of n is not prescribed.

Theorem 1.1 (All-exponent nearest-gap classification). *For $q \geq 1$, $\sigma \in \{1, -1\}$, and positive n , the two conditions [equations \(2\)](#) and [\(4\)](#) hold exactly in the two rows*

q	σ	n	$g_{q,\sigma}$	$\Delta_{q,\sigma}/g_{q,\sigma}$
2	-1	2	6	λ^{-1}
4	1	6	40	λ

(7)

In particular, no odd exponent is a target. Every hypothetical target obeys the intrinsic selector

$$n \text{ odd} \implies 3 \mid q, \quad n \text{ even} \implies 3 \nmid q. \quad (8)$$

The boundary $q \geq 1$ is sharp: at $q = 0$ one has $D_{0,\sigma} = \sqrt{2}$ and $\lambda - D_{0,\sigma}^2 = \lambda^{-1}$.

The proof is an exact, finite certificate built on an unbounded argument. The parity selector first divides the proof into two genuinely different branches. For even n , algebraic reconstruction turns the coefficient content into a primitive factor rectangle. This gives the missing height separation $h < n < 2q$ for the negative unit exponent $-h$. Two linear forms in logarithms then give $h = O(\log q)$ and $q < 10^{32}$; certified rational-interval reductions yield $h < 192$ and $q < 90$, after which exact recurrence arithmetic is exhaustive. For odd n , the selector forces $3 \mid q$. Its factorisation has no analogue of $h < n$, so the proof must be reordered: a fixed-coefficient logarithmic form first bounds $\min\{h, q \log_\lambda \varphi\}$, and only then can the moving-coefficient form bound q . The same certified reductions give $h < 48$ or $q < 96$, and the exact terminal list is empty.

The scarcity in [theorem 1.1](#) does not come from scarcity of windows. [Theorem 2.3](#) proves that both even candidate windows occur for natural density $\log_\lambda \sqrt{2} - \eta = 0.3300\dots$, and that the density statement persists on every arithmetic progression. The exact norm-content condition, not the Archimedean approximation, is the rigid step.

Our second main theorem packages the classification as a complete arithmetic orbit result. Let

$$\mathcal{S}(u, v) = (v, u + 2v), \quad \mathcal{H}(u, v) = u^2 + v^2, \quad z_{q,\sigma} = (F_{q+1}, F_{q+1} + \sigma F_q), \quad (9)$$

and write $\lambda^r = C_r + P_r\sqrt{2}$. We call d the *anchor*: it is the Pell index at time $t = 0$ in the orbit equation below.

Theorem 1.2 (Complete Fibonacci-core Pell-orbit classification). *For every $q \geq 1$, $\sigma \in \{1, -1\}$, positive anchor d , and $t \geq 0$, one has*

$$\mathcal{H}(S^t z_{q,\sigma}) = P_{d+2t} \quad (10)$$

if and only if one of the following alternatives holds:

σ	q	d	t
1	1	3	arbitrary $t \geq 0$
-1	1	1	0
-1	2	3	0
-1	4	5	0

(11)

There is no hit at a positive even anchor. In particular, the consecutive-Fibonacci core $z_{q,1} = (F_{q+1}, F_{q+2})$ has no hit for $q \geq 2$.

The bridge behind this theorem is more general than the Fibonacci cores. For a positive ordered integral seed $z = (u, v)$, the map

$$A_z = (v - u) + u\sqrt{2} \quad (12)$$

obeys $A_{S^t z} = \lambda^t A_z$. Thus an integral linear dynamical system is integrally conjugate to multiplication by a quadratic unit. An orbit hit at anchor d is equivalent, outside the canonical unit seeds, to the anchored gap

$$\lambda^{d-1} - A_z^2 = g\lambda^{-(2t+1)}, \quad (13)$$

which automatically lies in the strict factor-two window. This equivalence retains the anchor and time in both directions; it is not merely a numerical analogy.

Several classical structures clarify why the proof works. The equations $N(X + Y\sqrt{2}) = \pm 1$ define the norm-one torus and its norm- -1 torsor, so coefficient content records the integral lattice multiple of a primitive torsor point. The matrix of S lies in $\mathrm{GL}_2(\mathbb{Z})$, and its symmetric-square action preserves the Lorentzian identity $2\mathcal{H}^2 - \mathcal{B}^2 - \mathcal{Q}^2 = 0$ used in the orbit bridge. The two involutions of the biquadratic field $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ have different jobs: one controls Pell norms, while the other separates the dominant and conjugate Fibonacci terms in the logarithmic forms. Finally, the local results combine p -adic rank lifting with an Archimedean irrational rotation. These are operational connections: each supplies an invariant, factorisation, height, or recurrence mechanism used below.

Interactions between the Fibonacci and Pell sequences have been studied from several directions. Alekseyev determined the exact intersection of the two sequences [2], and Bravo, Herrera and Luca extended the question to common values of their generalised analogues [5]. Pomeo and Bravo measured how closely a Pell number can approach a Fibonacci number [15], Luca and Zottor classified the Fibonacci numbers occurring among the Y -coordinates of Pell equations [13], and Kafle, Luca, Montejano, Szalay and Togbé treated X -coordinates that are products of two Fibonacci numbers [10]. Each of those questions asks for a value coincidence between two sequences, or for an Archimedean bound on their difference. The present question is different in kind. The window equation (2) is only its Archimedean part, and the arithmetic lies in the exact norm-content equation (4) imposed on the gap itself. That equation sees both coefficient coordinates of $\Delta_{q,\sigma}$ at once, which is why a positive-density supply of windows leaves exactly two rows.

The nearest-gap problem arose in an ongoing programme on the Markoff–Frobenius uniqueness conjecture. A positive integral triple (a, b, c) with $a \leq b \leq c$ is a Markoff triple when

$$a^2 + b^2 + c^2 = 3abc, \quad (14)$$

and the conjecture asks whether the largest entry c determines the unordered pair $\{a, b\}$. The problem has arithmetic, geometric, word- combinatorial, and trace formulations; see Aigner [1], the integral-necklace model of Fisac [8], and Reutenauer’s account of Christoffel words and Markoff numbers [16]. The present paper neither proves nor disproves this conjecture. In particular, we do not prove that a hypothetical Markoff collision must enter the Fibonacci-core family in [theorem 1.2](#). The arithmetic theorem is complete on its stated domain; the upstream reduction remains open.

The local theory also draws a sharp methodological boundary. A natural plus-sign cross-gcd on genuine nearest windows contains arbitrary prescribed powers of 7 and 17, and more generally simultaneous products from an infinite class of split primes. A prime-rank parity criterion is exact, while the split prime 241 gives a sharp obstruction. These facts rule out any proof based only on uniform boundedness, squarefreeness, or finitely many compatible local rank clocks. They do not produce a norm-content target.

[Figure 1](#) displays the dependency structure and the one unproved Markoff-facing arrow.

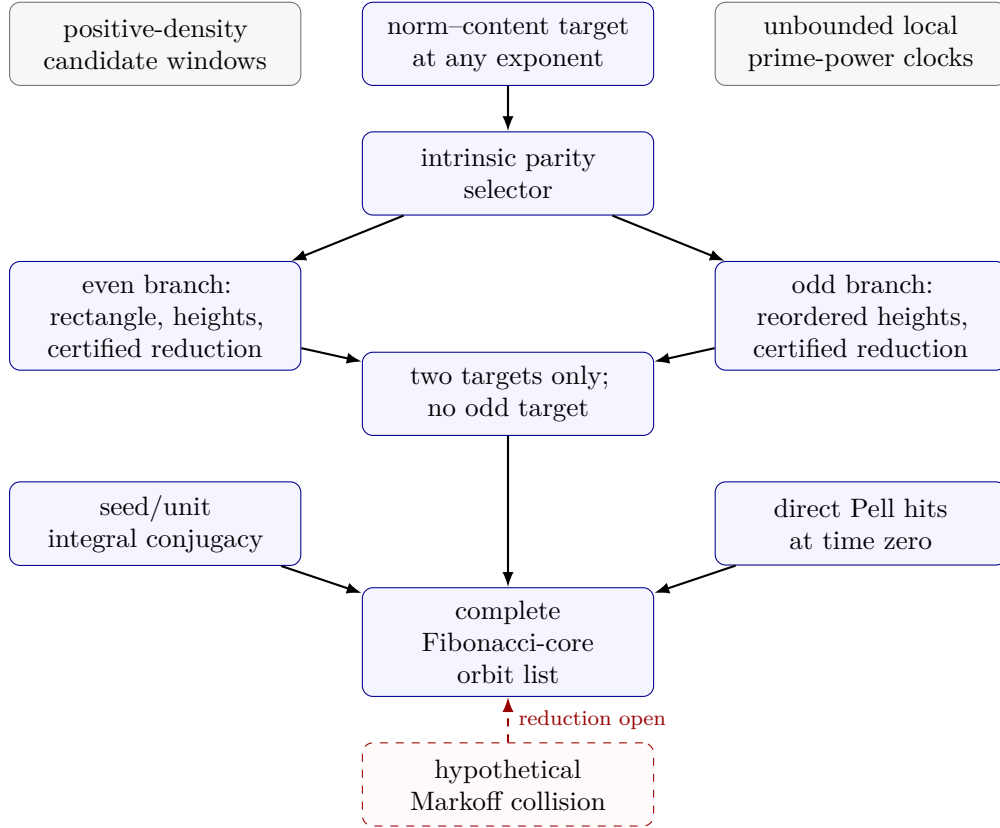


FIGURE 1. Logical architecture. Solid arrows are proof dependencies: the orbit list combines the independent seed/unit conjugacy, the all-exponent classification, and the direct time-zero Pell hits. Density and local clocks are contextual, not hypotheses. The dashed red arrow is the unproved Markoff-to-orbit reduction.

The paper is organised as follows. [Section 2](#) develops the quadratic-unit dictionary, the exact window densities, and the primitive factor rectangle. [Sections 3 to 5](#) prove

the even branch, and [section 6](#) proves the odd exclusion. [Section 7](#) gives the general seed/unit integral conjugacy and the complete Fibonacci-core orbit theorem. [Section 8](#) develops the local rank criterion and simultaneous prime-power recurrence. We close by stating the exact Markoff-facing boundary and further directions. [Section A](#) specifies the finite certificates and their coverage.

2. QUADRATIC UNITS AND THE INTRINSIC NEAREST-GAP SYSTEM

This section builds the arithmetic dictionary used throughout the paper. A candidate window is an Archimedean inequality; a target is a candidate whose two coefficient coordinates also satisfy the exact norm–content equation. Keeping those two levels separate is essential.

2.1. Quadratic integers and primitive-unit normalisation. We use the Fibonacci sequence

$$F_0 = 0, \quad F_1 = 1, \quad F_{r+2} = F_{r+1} + F_r, \quad (15)$$

and the Pell coordinates defined for every $r \in \mathbb{Z}$ by

$$\lambda^r = C_r + P_r\sqrt{2}, \quad \lambda = 1 + \sqrt{2}. \quad (16)$$

Thus $P_0 = 0$, $P_1 = 1$, $P_{r+2} = 2P_{r+1} + P_r$, and $C_r = P_r + P_{r-1}$ for $r \geq 1$. Since $\bar{\lambda} = 1 - \sqrt{2} = -\lambda^{-1}$,

$$C_r^2 - 2P_r^2 = (-1)^r. \quad (17)$$

Multiplication by λ^r acts on coefficient columns by the integral matrix

$$\begin{pmatrix} C_r & 2P_r \\ P_r & C_r \end{pmatrix}, \quad (18)$$

whose determinant is $(-1)^r$. It therefore preserves coefficient content. Equivalently, for $\alpha \in \mathbb{Z}[\sqrt{2}]$, $\text{cont}(\alpha)$ is the largest positive integer c such that $\alpha \in c\mathbb{Z}[\sqrt{2}]$, and multiplication by a unit is a unimodular change of the coefficient lattice.

There is also a useful arithmetic-geometric reading. The Pell conic $X^2 - 2Y^2 = 1$ is the norm-one torus for $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$, while $X^2 - 2Y^2 = -1$ is its norm- -1 torsor. The coefficient-primitive integral points on the latter with positive principal embedding are precisely λ^j for odd $j \in \mathbb{Z}$. Thus the target condition below says that the primitive direction of a gap lands on this torsor; see, for example, the Pell-conic viewpoint in [\[11, Chapter 8\]](#).

More explicitly, the regular representation

$$\iota(a + b\sqrt{2}) = \begin{pmatrix} a & 2b \\ b & a \end{pmatrix}, \quad \det \iota(a + b\sqrt{2}) = a^2 - 2b^2, \quad (19)$$

identifies the integral points of the norm-one torus with the rank-one arithmetic group

$$T(\mathbb{Z}) = \{\pm\lambda^{2m} : m \in \mathbb{Z}\}. \quad (20)$$

The norm- -1 integral fibre is the coset $\lambda T(\mathbb{Z})$. Hence a target gap is a rational lattice multiple of a point in this coset, and division by the coefficient content recovers its primitive arithmetic-group direction. This also explains why the matrix in [equation \(18\)](#) preserves content: it is the integral regular representation of a unit.

For the rest of the paper let

$$f = F_q, \quad x = F_{q+1}, \quad k = R_q = 2x^2 - f^2, \quad (21)$$

and define

$$A_q = f + x\sqrt{2}, \quad B_q = x\sqrt{2} - f. \quad (22)$$

For $q \geq 1$, both numbers are positive and $A_q B_q = k$. The two signs in [equation \(1\)](#) are simply $D_{q,1} = A_q$ and $D_{q,-1} = B_q$.

Lemma 2.1 (Uniqueness and unit normalisation). *For a fixed pair (q, σ) with $q \geq 1$, at most one positive integer n satisfies*

$$D_{q,\sigma}^2 < \lambda^n < 2D_{q,\sigma}^2. \quad (23)$$

If this integer exists, put $\Delta = \lambda^n - D_{q,\sigma}^2$ and $g = \text{cont}(\Delta)$. Then

$$N(\Delta) = -g^2 \quad (24)$$

if and only if

$$\Delta = g\lambda^j \quad (25)$$

for a unique odd integer j . In that case $M = n - j$ is positive and $M \equiv n - 1 \pmod{2}$.

Proof. The ratio of two distinct integral powers of λ is at least $\lambda > 2$, proving uniqueness. If [equation \(24\)](#) holds, then Δ/g is a coefficient-primitive algebraic integer of norm -1 and positive principal embedding. The unit group of $\mathbb{Z}[\sqrt{2}]$ gives $\Delta/g = \lambda^j$ for one odd j . The converse is immediate from [equation \(17\)](#). Finally, $\lambda^j = \Delta/g < \lambda^n$, so $j < n$ and $M = n - j > 0$; the congruence for M follows because j is odd. $\square$

Proposition 2.2 (Global parity selector). *Every target satisfies*

$$n \text{ odd} \implies 3 \mid q, \quad n \text{ even} \implies 3 \nmid q. \quad (26)$$

Proof. The coefficient pair of the gap is

$$(C_n - (f^2 + 2x^2), P_n - 2\sigma f x). \quad (27)$$

Modulo 2, one has $C_n \equiv 1$, $P_n \equiv n$, and $f^2 + 2x^2 \equiv f$. If the target condition holds, the quotient by the content is an odd power of λ , whose two coefficients are odd. Consequently the two entries in [equation \(27\)](#) have the same parity, and

$$n \equiv 1 - F_q \pmod{2}. \quad (28)$$

The Fibonacci number F_q is even exactly when $3 \mid q$, proving [equation \(26\)](#). $\square$

The selector is a conclusion, not an external admissibility hypothesis. The even branch automatically has $3 \nmid q$, hence f and k are odd and M is odd. Those are precisely the hypotheses used by the primitive reconstruction below. The odd branch has a different factorisation and will be treated in [section 6](#).

2.2. Candidate windows are abundant. Before imposing the norm-content equation, it is useful to know how often the Archimedean window itself occurs. This separates the elementary supply of candidates from the arithmetic rigidity that eventually leaves only two targets. Put

$$\vartheta = \log_\lambda \sqrt{2}, \quad \eta = \log_\lambda \left(\frac{\sqrt{2}\varphi + 1}{\lambda(\sqrt{2}\varphi - 1)} \right), \quad \varphi = \frac{1 + \sqrt{5}}{2}. \quad (29)$$

Theorem 2.3 (Exact density of the even candidate windows). *Call the sign σ available at q if an even positive integer n satisfies [equation \(23\)](#). Then*

$$0 < \eta < \vartheta < \frac{1}{2}, \quad (30)$$

and the four sign patterns have natural densities

available signs	both	plus only	minus only	neither
density	$\vartheta - \eta$	η	η	$1 - \vartheta - \eta$

(31)

The same relative densities hold after restricting q to any fixed arithmetic progression. In particular, they remain valid on either class $q \equiv 1, 2 \pmod{3}$ and on their union $3 \nmid q$.

Proof. Write $x_\sigma(q) = \log_\lambda D_{q,\sigma}$. If $n = 2m$, then [equation \(23\)](#) is equivalent, away from its strict endpoints, to

$$\{x_\sigma(q)\} \in I, \quad I = (1 - \vartheta, 1). \quad (32)$$

Binet's formula gives

$$x_\sigma(q) = q\alpha + \beta_\sigma + O(\varphi^{-2q}), \quad \alpha = \frac{\log \varphi}{\log \lambda}, \quad \beta_\sigma = \log_\lambda \left(\frac{\sqrt{2}\varphi + \sigma}{\sqrt{5}} \right). \quad (33)$$

The number α is irrational. Indeed, a rational relation would put equal positive powers of φ and λ in $\mathbb{Q}(\sqrt{5}) \cap \mathbb{Q}(\sqrt{2}) = \mathbb{Q}$, whereas no positive power of φ is rational. Hence $q\alpha + \beta_-$ is equidistributed modulo one by Weyl's theorem [19]. Equivalently, its empirical measures converge to Haar probability measure on $\mathbb{R}/\mathbb{Z}$. Since

$$\beta_+ - \beta_- = 1 + \eta, \quad (34)$$

the minus interval is $I_- = (1 - \vartheta, 1)$ and the plus interval, expressed in the same phase coordinate, is

$$I_+ = (1 - \vartheta - \eta, 1 - \eta). \quad (35)$$

The endpoint order in [figure 2](#) gives the four lengths in [equation \(31\)](#); these lengths are exactly the Haar measures of the corresponding phase regions. The inequalities in [equation \(30\)](#) reduce respectively to $1 + \sqrt{2} > \sqrt{5}$, $\varphi > 3/2$, and $\lambda > 2$. The exponentially decaying error in [equation \(33\)](#) can change membership only near the four endpoints, a set of density zero. On a progression $q = q_0 + Q\ell$, the increment $Q\alpha$ remains irrational, so the same argument applies. $\square$

There is an exact probability interpretation of this deterministic theorem. If Q_N is chosen uniformly from $\{1, \dots, N\}$, then the four availability events converge to the probabilities in [equation \(31\)](#). The probability space is simply the phase circle with Haar measure; no independence assumption or random mechanism enters the arithmetic.

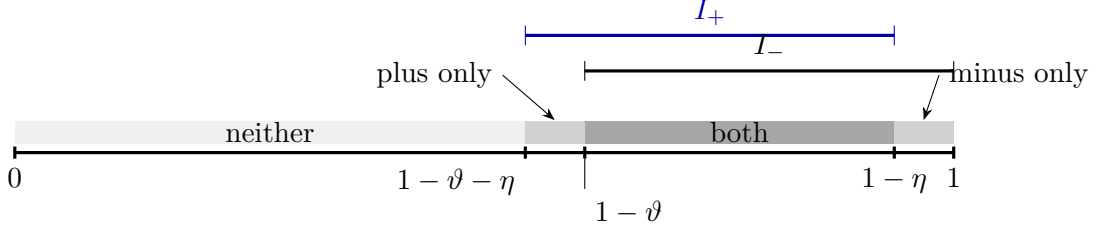


FIGURE 2. The exact phase partition for the two even nearest-window candidates. Irrational rotation distributes the limiting phase uniformly; the four natural densities are therefore the displayed interval lengths. Availability is only an Archimedean condition and does not imply the exact norm-content equation.

Numerically, $\vartheta = 0.3932198506\dots$ and $\eta = 0.0631959999\dots$, so both windows occur for about 33% of all indices. The final classification will show that this positive-density supply contains only two norm-content targets. Scarcity of candidate windows is therefore not the exclusion mechanism.

2.3. Primitive reconstruction in the even branch. The next reconstruction is the geometric input behind the later height separation. We include the proof because using an arbitrary divisor of $(k^2 - 1)/4$ here would be invalid.

Lemma 2.4 (Primitive rectangle reconstructed from a target). *Assume that n is even and that [equations \(23\)](#) and [\(24\)](#) hold. With $M = n - j$, there is an odd integer W such that*

$$W^2 = C_M^2 + 1 - k^2, \quad \gcd(k, W) = 1, \quad |W| < C_M. \quad (36)$$

Put

$$U = \frac{C_M - W}{2}, \quad V = \frac{C_M + W}{2}. \quad (37)$$

Then U, V are positive integers of opposite parity and

$$UV = \frac{k^2 - 1}{4}. \quad (38)$$

Moreover, $g/2$ equals U on the plus branch and V on the minus branch, and

$$g < k, \quad U \geq \sqrt{k}, \quad V \geq \sqrt{k}. \quad (39)$$

Proof. On the plus branch, division of $\lambda^n - A_q^2 = g\lambda^j$ by λ^j gives

$$A_q^2 \lambda^{-j} = C_M - g + P_M \sqrt{2}. \quad (40)$$

On the minus branch the same calculation gives

$$B_q^2 \lambda^{-j} = C_M - g + P_M \sqrt{2}. \quad (41)$$

Define $W = C_M - g$ in [equation \(40\)](#) and $W = g - C_M$ in [equation \(41\)](#); thus the rational coefficient in those two equations is, respectively, W and $-W$. The upper half of the strict window gives $D_{q,\sigma}^2 > \lambda^n/2$, and therefore

$$g\lambda^j = \lambda^n - D_{q,\sigma}^2 < \frac{\lambda^n}{2}.$$

Thus $2g < \lambda^{n-j} = \lambda^M$. Since M is odd,

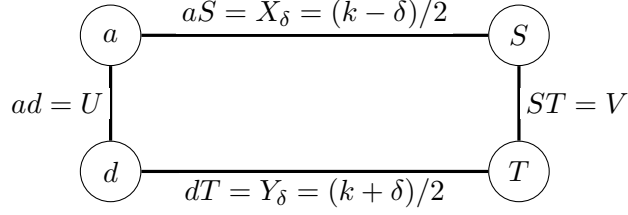


FIGURE 3. The primitive factor rectangle. Coprimality of the consecutive horizontal products allocates every prime power to one corner in each row. The opposite-corner quadratic factors $S^2 + d^2$ and $(T^2 + a^2)/2$ are coprime by [equation \(47\)](#), allowing Lagrange's identity to force two conics.

$$\frac{\lambda^M}{2} = C_M + \frac{\lambda^{-M}}{2} < C_M + 1. \quad (42)$$

Thus $g \leq C_M$. Equality would give $W = 0$ and $k^2 = C_M^2 + 1$, which is impossible modulo 8 because k and C_M are odd. Therefore [equation \(36\)](#) has $|W| = C_M - g < C_M$.

Taking norms in [equations \(40\)](#) and [\(41\)](#) and using [equation \(17\)](#) proves the first equality in [equation \(36\)](#). To prove primitivity, suppose that an odd prime p divides both k and W . The norm relation then gives $p \mid P_M$. After multiplication by a Pell unit, p would divide both coefficients of A_q^2 or B_q^2 . Those coefficients are $f^2 + 2x^2$ and $2fx$ up to signs. Since $\gcd(f, x) = 1$ and $p \mid 2x^2 - f^2$, this is impossible. Hence $\gcd(k, W) = 1$.

Factoring $C_M^2 - W^2 = k^2 - 1$ proves [equation \(38\)](#). Since M and k are odd, the norm relation shows that W is odd. Hence $U + V = C_M$ is odd, so U and V have opposite parities. Put

$$X_\delta = \frac{k - \delta}{2}, \quad Y_\delta = \frac{k + \delta}{2}. \quad (43)$$

The two factors X_δ and Y_δ are coprime and consecutive. Choose the unique $\delta \in \{1, -1\}$ for which Y_δ is even when U is even, and X_δ is even when V is even. Define

$$a = \gcd(U, X_\delta), \quad d = \gcd(U, Y_\delta), \quad S = \frac{X_\delta}{a}, \quad T = \frac{Y_\delta}{d}. \quad (44)$$

Because $X_\delta Y_\delta = UV$ and $\gcd(X_\delta, Y_\delta) = 1$, every prime power in U occurs wholly in exactly one of X_δ, Y_δ . Therefore $\gcd(U, X_\delta) \gcd(U, Y_\delta) = U$, and [equation \(44\)](#) gives

$$U = ad, \quad V = ST, \quad aS = X_\delta, \quad dT = Y_\delta, \quad dT - aS = \delta. \quad (45)$$

The same prime-support separation gives

$$\gcd(a, d) = \gcd(a, T) = \gcd(S, d) = \gcd(S, T) = 1. \quad (46)$$

Our choice of δ makes a, T odd and makes S, d of opposite parity. We claim that

$$\gcd\left(S^2 + d^2, \frac{T^2 + a^2}{2}\right) = 1. \quad (47)$$

Both factors are odd. If an odd prime p divided them, then none of a, d, S, T would vanish modulo p . Put $r = S/d$ and $s = T/a$ in $\mathbb{F}_p$. The two norm congruences give $r^2 = s^2 = -1$. The relation $dT - aS = \delta$ shows that $r \neq s$, so $s = -r$. It follows that $k = aS + dT = ad(r + s)$ and $W = ST - ad = ad(rs - 1)$ both vanish modulo p , contrary to $\gcd(k, W) = 1$. This proves [equation \(47\)](#).

We therefore have

$$S^2 + d^2 = H^2, \quad T^2 + a^2 = 2J^2 \quad (48)$$

for coprime positive integers H, J . One direct way to see the square conclusion is Lagrange's identity:

$$(S^2 + d^2) \frac{T^2 + a^2}{2} = \frac{(ad + ST)^2 + (dT - aS)^2}{2} = P_M^2. \quad (49)$$

The two factors on the left are coprime by [equation \(47\)](#), so each is a square.

For the positive Pythagorean pair (S, d) , integrality gives $H - S \geq 1$ and $H - d \geq 1$. Consequently

$$d^2 = (H - S)(H + S) \geq 2S + 1, \quad S^2 = (H - d)(H + d) \geq 2d + 1, \quad (50)$$

or equivalently

$$2S \leq d^2 - 1, \quad 2d \leq S^2 - 1. \quad (51)$$

Using $k = 2aS + \delta = 2dT - \delta$, the preceding inequalities, and positivity gives

$$k \leq a(d^2 - 1) + 1 \leq ad^2 \leq (ad)^2 = U^2, \quad (52)$$

$$k \leq T(S^2 - 1) + 1 \leq TS^2 \leq (ST)^2 = V^2. \quad (53)$$

These inequalities prove the two factor bounds in [equation \(39\)](#). Finally, the definitions of W show $g = C_M - W = 2U$ on the plus branch and $g = C_M + W = 2V$ on the minus branch. In each case the selected factor is the smaller of U, V , so $g < 2\sqrt{UV} = \sqrt{k^2 - 1} < k$. $\square$

Remark 2.5. The rectangle is reconstructed from the full coefficient-content target, not from a numerical near equality. This is why the lower bound [equation \(39\)](#) is available. The proof of the classification would fail if $g/2$ were replaced by an arbitrary divisor of $(k^2 - 1)/4$.

With the arithmetic dictionary and the primitive rectangle in hand, we now enter the even branch. We first determine which primitive-unit exponents can lead, and then separate those exponents quantitatively from the nearest Pell exponent.

3. LEADING UNIT EXPONENTS IN THE EVEN BRANCH

Assume throughout [sections 3 to 5](#) that an even-exponent target is fixed, and retain j and $M = n - j$ from [lemma 2.1](#). By [proposition 2.2](#), $3 \nmid q$ and M is odd; in particular, [lemma 2.4](#) applies. The first task is to show that every target in the unresolved range has a genuinely negative normalised exponent.

Lemma 3.1 (Ratio bounds). *For $q \geq 3$,*

$$\lambda B_q < A_q < 3B_q. \quad (54)$$

Proof. Using $F_{q+1} = F_q + F_{q-1}$, direct expansion gives

$$A_q - \lambda B_q = \sqrt{2}F_q - 2F_{q-1} > 0, \quad (55)$$

because $F_q/F_{q-1} \geq 3/2 > \sqrt{2}$ for $q \geq 3$. Similarly,

$$3B_q - A_q = 2(\sqrt{2}F_{q+1} - 2F_q) > 0. \quad (56)$$

Indeed, $F_{q+1}/F_q \geq 3/2 > \sqrt{2}$ for $q \geq 3$, which proves the last inequality. $\square$

Proposition 3.2 (Negative-exponent reduction). *Every even-exponent target with $q \geq 7$ has*

$$j = -h, \quad h \geq 3 \text{ odd.} \quad (57)$$

Proof. On the plus branch, equation (40) has positive rational coefficient $W = C_M - g$. Quadratic conjugation gives

$$2W = A_q^2 \lambda^{-j} - B_q^2 \lambda^j > 0. \quad (58)$$

Thus $\lambda^j < A_q/B_q < 3 < \lambda^2$. Since j is odd, $j \leq 1$. On the minus branch the actual rational coefficient is $C_M - g = -W > 0$. Conjugating equation (41) gives

$$2(C_M - g) = B_q^2 \lambda^{-j} - A_q^2 \lambda^j > 0, \quad (59)$$

and hence

$$\lambda^j < \frac{B_q}{A_q} < \lambda^{-1}. \quad (60)$$

Hence the odd integer j is at most -3 there.

It remains to remove $j = 1$ and $j = -1$ on the plus branch. If $j = 1$, comparison of the $\sqrt{2}$ -coefficients in equation (40) gives

$$P_M = F_{q-1}^2 + F_{q+1}^2 = 3F_q^2 + 2(-1)^q. \quad (61)$$

This recurrence equation has exactly

$$(q, M) = (1, 1), (2, 3), (4, 5) \quad (62)$$

among nonnegative q and positive odd M . This follows directly from the complete near-square classification of Alekseyev and Tengely [3, Table 1]. In their notation the Pell sequence is $U_M(2, -1)$. They prove that all Pell values of the forms $3y^2 + 2$ and $3y^2 - 2$ are, respectively,

$$\{2, 5, 29\} \quad \text{and} \quad \{1\}. \quad (63)$$

If q is even, equation (61) has the first form in equation (63). The value $2 = P_2$ is removed because M is odd, while $5 = P_3$ and $29 = P_5$ give $F_q = 1$ and $F_q = 3$. Even parity of q leaves $(q, M) = (2, 3), (4, 5)$. If q is odd, the second form in equation (63) gives $P_M = 1$, hence $M = 1$ and $F_q = 1$; odd parity leaves $(q, M) = (1, 1)$. Direct substitution proves the converse.

The nearest plus window is absent at $q = 2$, and $q = 4$ is the low plus target in equation (7). Thus no $j = 1$ target has $q \geq 7$.

If $j = -1$, the radical coefficient instead becomes

$$P_M = F_q^2 + 2F_{q+1}^2 + 2F_q F_{q+1} = F_{q+1}^2 + F_{q+2}^2 = F_{2q+3}. \quad (64)$$

Alekseyev's complete Fibonacci–Pell intersection theorem states that the common values of the two sequences are $0, 1, 2, 5$ [2, Theorem 9]. Since $2q + 3 \geq 17$ for $q \geq 7$, equation (64) is impossible. All remaining exponents therefore have the form equation (57). $\square$

Remark 3.3 (Independent verification of the near-square step). The published theorem [3] is the completeness input; this boundary classification is not new here. As an independent audit, the negative-Pell identity converts equation (61), with $y = F_q$ and $s = (-1)^q$, into

$$C_M^2 = 18y^4 + 24sy^2 + 7. \quad (65)$$

After $y = z + 1$, two complete Magma V2.29-9 integral-point computations give exactly the points corresponding to [equation \(62\)](#). The supplement contains both inputs, complete transcripts, and independent arithmetic verifiers. These computations corroborate the published theorem and are not used as an unrecorded bounded search.

Thus every even-exponent target in the unresolved range has primitive unit direction λ^{-h} with $h \geq 3$ odd. The qualitative leading-unit ambiguity has disappeared; the next task is quantitative, namely to separate the negative exponent h from the nearest Pell exponent n .

4. HEIGHT SEPARATION IN THE EVEN BRANCH

Continue under the standing hypothesis of an even-exponent target with $q \geq 7$. By [proposition 3.2](#),

$$\lambda^n - D_{q,\sigma}^2 = g\lambda^{-h}, \quad h \geq 3 \text{ odd}. \quad (66)$$

We first turn the reconstructed rectangle into a separation between h and n .

Lemma 4.1 (Divisor-height separation). *Every target in [equation \(66\)](#) satisfies*

$$\begin{aligned} \sigma = 1 : \quad & h \leq n - 5, \\ \sigma = -1 : \quad & h \leq n - 3. \end{aligned} \quad (67)$$

In particular, $h < n$.

Proof. Put

$$L = f^2 + 2x^2. \quad (68)$$

Since $\lambda^{-h} = -C_h + P_h\sqrt{2}$ for odd h , coefficient comparison in [equation \(66\)](#) gives

$$gC_h = L - C_n, \quad gP_h = P_n - 2\sigma fx. \quad (69)$$

Let $t = g/2$. By [lemma 2.4](#), $t \geq \sqrt{k}$. For every positive odd h ,

$$C_h = \frac{\lambda^h - \lambda^{-h}}{2} > \frac{\lambda^h}{3}. \quad (70)$$

On the plus branch, $L < A_q^2 < \lambda^n$. Hence

$$t = \frac{L - C_n}{2C_h} < \frac{3}{2}\lambda^{n-h}. \quad (71)$$

On the minus branch, [equation \(54\)](#) gives

$$L = \frac{A_q^2 + B_q^2}{2} < 5B_q^2 < 5\lambda^n, \quad (72)$$

and therefore $t < (15/2)\lambda^{n-h}$. Combining these inequalities with $t \geq \sqrt{k}$ yields

$$\lambda^{2(n-h)} > \begin{cases} 4k/9, & \sigma = 1, \\ 4k/225, & \sigma = -1. \end{cases} \quad (73)$$

The sequence R_q is increasing for $q \geq 3$: with $f = F_q$ and $x = F_{q+1} < 2f$,

$$R_{q+1} - R_q = 3f^2 + 4fx - x^2 > 0. \quad (74)$$

Hence $k \geq R_7 = 713$, and $n - h$ is odd. Direct comparison in [equation \(73\)](#) excludes $n - h \leq 3$ in the plus case and $n - h \leq 1$ in the minus case. This proves [equation \(67\)](#). $\square$

We shall use the following explicit specialization of Matveev's theorem [14, Corollary 2.3]. For a positive algebraic number α , write $h_{\text{abs}}(\alpha)$ for its absolute logarithmic Weil height.

Lemma 4.2 (Matveev, three positive real logarithms). *Let K be a real number field of degree D , let $\alpha_1, \alpha_2, \alpha_3 \in K$ be positive, and let*

$$\Lambda = b_1 \log \alpha_1 + b_2 \log \alpha_2 + b_3 \log \alpha_3 \neq 0 \quad (75)$$

with $b_i \in \mathbb{Z}$. If

$$A_i \geq \max\{Dh_{\text{abs}}(\alpha_i), |\log \alpha_i|, 0.16\} \quad (76)$$

and $B \geq \max_i |b_i|$, then

$$\log |\Lambda| > -1.4 \cdot 30^6 3^{9/2} D^2 (1 + \log D) A_1 A_2 A_3 (1 + \log B). \quad (77)$$

The first application reduces the a priori linear range for h to a logarithmic range.

Proposition 4.3 (Logarithmic bound for the negative exponent). *Every target with $q \geq 7$ satisfies*

$$h < 2.4 \cdot 10^{14} (1 + \log(2q)) \quad (78)$$

and

$$n < 2q. \quad (79)$$

Proof. Put

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \epsilon = (-1)^q, \quad u = \sigma + \sqrt{2}\varphi, \quad v = -\sigma + \sqrt{2}\varphi^{-1}. \quad (80)$$

Binet's formula gives

$$D_{q,\sigma} = \frac{u\varphi^q + \epsilon v\varphi^{-q}}{\sqrt{5}}. \quad (81)$$

Set

$$\delta = \epsilon \frac{v}{u} \varphi^{-2q}, \quad A_0 = \frac{u^2}{5} \varphi^{2q}. \quad (82)$$

Then $D_{q,\sigma}^2 = A_0(1 + \delta)^2$. The elementary bounds $1 < u < 4$ and $|v/u| < 2$ give

$$|\delta| < 2 \left(\frac{2}{3} \right)^{14} < \frac{1}{100}. \quad (83)$$

By lemma 4.1 and the nearest window,

$$\lambda^{h+3} < 2 \frac{16}{5} \left(\frac{101}{100} \right)^2 \varphi^{2q} < 7 \varphi^{2q}. \quad (84)$$

Since $\lambda > 12/5$, this strengthens equation (83) to

$$|\delta| < \frac{1750}{1728} \lambda^{-h}. \quad (85)$$

By lemma 2.4, $g < k = A_q B_q$. Therefore $g/D_{q,\sigma}^2 < 3$ for either sign, by equation (54). Define

$$\Lambda_0 = n \log \lambda - 2q \log \varphi - \log \left(\frac{u^2}{5} \right). \quad (86)$$

After dividing [equation \(66\)](#) by A_0 , the estimates [equations \(83\)](#) and [\(85\)](#) give

$$|e^{\Lambda_0} - 1| < \frac{128}{25} \lambda^{-h} < \frac{10}{27}. \quad (87)$$

The elementary logarithm estimate on this interval yields

$$0 < |\Lambda_0| < 9\lambda^{-h}. \quad (88)$$

The nonvanishing is exact. In $K = \mathbb{Q}(\sqrt{2}, \sqrt{5})$ one has $N_{K/\mathbb{Q}}(u) = -1$ and $N_{K/\mathbb{Q}}(\lambda) = N_{K/\mathbb{Q}}(\varphi) = 1$. Equality in [equation \(86\)](#) would give $1 = 5^{-4}$ after taking full norms.

We next verify [equation \(79\)](#). From [equation \(81\)](#), $D_{q,\sigma} < 4\varphi^q$. Also $\lambda/\varphi > 7/5$ and $(7/5)^{14} > 32$. Thus, for $q \geq 7$,

$$\lambda^{2q} > 32\varphi^{2q} > 2D_{q,\sigma}^2 > \lambda^n. \quad (89)$$

For [lemma 4.2](#), the field degree is 4 and admissible parameters for [equation \(86\)](#) are

$$A_1 = 2 \log \lambda, \quad A_2 = 2 \log \varphi, \quad A_3 = 20, \quad B = 2q. \quad (90)$$

Indeed,

$$h_{\text{abs}}(u^2/5) \leq 3 \log 2 + \log \varphi + \log 5 < 5, \quad (91)$$

and $1/5 < u^2/5 < 16/5$. The constant before A_3 satisfies the exact outward inequality

$$1.4 \cdot 30^6 3^{9/2} 4^2 (1 + \log 4) (2 \log \lambda) (2 \log \varphi) < 9.275 \cdot 10^{12}. \quad (92)$$

Writing $H = 1 + \log(2q)$, comparison with [equation \(88\)](#) gives

$$h \log \lambda < 1.855 \cdot 10^{14} H + \log 9. \quad (93)$$

Finally $\log \lambda > 4/5$. Rounding only upward proves [equation \(78\)](#). $\square$

The fixed-coefficient form has therefore done two jobs: it makes h logarithmic in q and keeps n below $2q$. It does not yet bound q . The next section supplies that missing absolute bound by allowing the third logarithmic coefficient to depend on the now-controlled exponent h .

5. EFFECTIVE CLASSIFICATION OF THE EVEN BRANCH

We now bound q , reduce both exponents by exact rational intervals, and finish the even-exponent branch of [theorem 1.1](#). Until the terminal enumeration, we retain the standing hypothesis that an even-exponent target exists with $q \geq 7$.

5.1. A moving-coefficient logarithmic form. Quadratic conjugation of [equation \(66\)](#), followed by elimination of g , gives the exact trace identity

$$\lambda^n + \lambda^{-n-2h} = D_{q,\sigma}^2 + \overline{D}_{q,\sigma}^2 \lambda^{-2h}. \quad (94)$$

Retain $\epsilon = (-1)^q$ and put

$$\begin{aligned} u_\sigma &= \sigma + \sqrt{2}\varphi, & u'_\sigma &= -\sigma + \sqrt{2}\varphi^{-1}, \\ v_\sigma &= \sigma - \sqrt{2}\varphi, & v'_\sigma &= -\sigma - \sqrt{2}\varphi^{-1}. \end{aligned} \quad (95)$$

Binet's formula becomes

$$\begin{aligned} D_{q,\sigma} &= \frac{u_\sigma \varphi^q + \epsilon u'_\sigma \varphi^{-q}}{\sqrt{5}}, \\ \overline{D}_{q,\sigma} &= \frac{v_\sigma \varphi^q + \epsilon v'_\sigma \varphi^{-q}}{\sqrt{5}}. \end{aligned} \quad (96)$$

The key is to retain both growing φ^{2q} terms inside one coefficient:

$$\mathcal{A}_{\sigma,h} = \frac{u_\sigma^2 + v_\sigma^2 \lambda^{-2h}}{5}. \quad (97)$$

Substitution in [equation \(94\)](#) gives

$$\lambda^n = \mathcal{A}_{\sigma,h} \varphi^{2q} + \mathcal{R}_{\sigma,q,h}, \quad (98)$$

where

$$\begin{aligned} \mathcal{R}_{\sigma,q,h} &= \frac{2\epsilon}{5} \left((1 - \sigma\sqrt{2}) + (1 + \sigma\sqrt{2})\lambda^{-2h} \right) \\ &\quad + \frac{\varphi^{-2q}}{5} \left((u'_\sigma)^2 + (v'_\sigma)^2 \lambda^{-2h} \right) - \lambda^{-n-2h}. \end{aligned} \quad (99)$$

Here we used $u_\sigma u'_\sigma = 1 - \sigma\sqrt{2}$ and $v_\sigma v'_\sigma = 1 + \sigma\sqrt{2}$. Direct coefficient estimates give

$$|\mathcal{R}_{\sigma,q,h}| < 4, \quad \frac{1}{5} < \mathcal{A}_{\sigma,h} < 7. \quad (100)$$

Since $\varphi^{14} = 377\varphi + 233 > 1597/2$, normalising [equation \(98\)](#) and using [equation \(100\)](#) yields the logarithmic form

$$\Lambda_1 = n \log \lambda - 2q \log \varphi - \log \mathcal{A}_{\sigma,h} \quad (101)$$

with

$$0 < |\Lambda_1| < 21\varphi^{-2q}. \quad (102)$$

The strict nonvanishing and the height of the moving coefficient must be controlled uniformly. From $\lambda^h = C_h + P_h\sqrt{2}$ one obtains

$$\mathcal{A}_{\sigma,h} = \lambda^{-h} \frac{2\sqrt{2}}{5} \left((4 + \sqrt{5})P_h + 2\sigma\varphi C_h \right). \quad (103)$$

Taking the full norm in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ gives

$$N_{K/\mathbb{Q}}(\mathcal{A}_{\sigma,h}) = \frac{64}{625} \left(4 + 3P_h^2 - 2\sigma C_h P_h \right)^2. \quad (104)$$

If $\Lambda_1 = 0$, full norms would force $64I^2 = 625$ for an integer I , which is impossible. We now record the height calculation. The inequalities $h_{\text{abs}}(\alpha\beta) \leq h_{\text{abs}}(\alpha) + h_{\text{abs}}(\beta)$ and $h_{\text{abs}}(\alpha + \beta) \leq h_{\text{abs}}(\alpha) + h_{\text{abs}}(\beta) + \log 2$ give

$$h_{\text{abs}}(u_\sigma), h_{\text{abs}}(v_\sigma) \leq \log 2 + \frac{\log 2}{2} + \frac{\log \varphi}{2} < \frac{3}{2}. \quad (105)$$

Since $h_{\text{abs}}(\lambda) = \frac{1}{2} \log \lambda$, the definition [equation \(97\)](#) yields

$$\begin{aligned} h_{\text{abs}}(\mathcal{A}_{\sigma,h}) &\leq 2h_{\text{abs}}(u_\sigma) + 2h_{\text{abs}}(v_\sigma) + 2h_{\text{abs}}(\lambda) + \log 10 \\ &< h + 9 < h + 11. \end{aligned} \quad (106)$$

Together with [equation \(100\)](#), this proves

$$h_{\text{abs}}(\mathcal{A}_{\sigma,h}) < h + 11, \quad |\log \mathcal{A}_{\sigma,h}| < 2. \quad (107)$$

Thus [lemma 4.2](#) applies to [equation \(101\)](#) with

$$A_1 = 2 \log \lambda, \quad A_2 = 2 \log \varphi, \quad A_3 = 4h + 44, \quad B = 2q. \quad (108)$$

Writing again $H = 1 + \log(2q)$ and using [equation \(92\)](#), comparison with [equation \(102\)](#) gives

$$2q \log \varphi < 9.275 \cdot 10^{12} (4h + 44)H + \log 21. \quad (109)$$

By [equation \(78\)](#) and $\log \varphi > 2/5$,

$$q < \frac{5}{4} \left[9.275 \cdot 10^{12} \left(9.6 \cdot 10^{14} H + 44 \right) H + 4 \right]. \quad (110)$$

At $q_0 = 10^{32}$, one has $H(q_0) < 79$ and the right-hand side of [equation \(110\)](#) is less than $7 \cdot 10^{31}$. After division by q , every term on the right is decreasing for $q \geq q_0$. Consequently

$$q < 10^{32}. \quad (111)$$

The moving coefficient [equation \(97\)](#) is the central mechanism of the proof. Its error decays with q , while its height grows only linearly with h , which is already logarithmic in q .

5.2. An absolute-value reduction lemma. We use a symmetric form of the Baker–Davenport reduction, closely related to the lemma of Dujella and Pethő [7]. We include a short proof to avoid a hidden sign condition.

Lemma 5.1 (Absolute-value reduction). *Let $\gamma, \mu \in \mathbb{R}$, let $p \in \mathbb{Z}$, let $Q \in \mathbb{Z}_{>0}$, and let $M \geq 1$. Suppose that $|Q\gamma - p| < 1/2$ and that integers a, b satisfy*

$$0 < |a\gamma - b + \mu| < AB^{-w}, \quad 1 \leq a \leq M, \quad (112)$$

where $A > 0$, $B > 1$, and $w > 0$ are real. Put

$$\eta = \|Q\mu\| - M|Q\gamma - p|, \quad (113)$$

where $\|x\|$ is the distance from x to the nearest integer. If $\eta > 0$, then

$$w < \frac{\log(AQ/\eta)}{\log B}. \quad (114)$$

Proof. Write $\|x\|$ for the distance to the nearest integer. Since $ap - bQ \in \mathbb{Z}$, the triangle inequality on $\mathbb{R}/\mathbb{Z}$ gives

$$\begin{aligned} \|Q\mu\| &\leq \|Q\mu + a(Q\gamma - p)\| + a|Q\gamma - p| \\ &\leq Q|a\gamma - b + \mu| + M|Q\gamma - p|. \end{aligned} \quad (115)$$

The strict hypothesis therefore gives $\eta < AQB^{-w}$, which is equivalent to [equation \(114\)](#). $\square$

The point of this symmetric form is that no sign of the logarithmic form is assumed. We apply it first to the fixed shift, reducing h or q , and then to the moving shift, which supplies the absolute bound for q .

5.3. **Certified reductions of the two exponents.** Set

$$\gamma = \frac{\log \varphi}{\log \lambda}, \quad \mu_\sigma = \frac{\log(u_\sigma^2/5)}{\log \lambda}. \quad (116)$$

Put

$$\delta_\sigma = \epsilon \frac{u'_\sigma}{u_\sigma} \varphi^{-2q}, \quad \tilde{\delta}_\sigma = \epsilon \frac{v'_\sigma}{v_\sigma} \varphi^{-2q}. \quad (117)$$

Dividing the Binet expansion of [equation \(94\)](#) by $(u_\sigma^2/5)\varphi^{2q}$ gives the exact identity

$$\begin{aligned} e^{\Lambda_0} - 1 &= 2\delta_\sigma + \delta_\sigma^2 + \frac{v_\sigma^2}{u_\sigma^2}(1 + \tilde{\delta}_\sigma)^2 \lambda^{-2h} \\ &\quad - \frac{5}{u_\sigma^2} \lambda^{-n-2h} \varphi^{-2q}. \end{aligned} \quad (118)$$

For both signs, direct comparison gives

$$1 < u_\sigma < 4, \quad \left| \frac{u'_\sigma}{u_\sigma} \right| < \frac{3}{2}, \quad \left| \frac{v_\sigma}{u_\sigma} \right| < 3, \quad \left| \frac{v'_\sigma}{v_\sigma} \right| < \frac{3}{2}. \quad (119)$$

Since $q \geq 7$, $\varphi^{14} > 1597/2$, so both delta terms have absolute value below $1/100$. The three terms in [equation \(118\)](#) are therefore bounded, respectively, by $4\varphi^{-2q}$, $10\lambda^{-2h}$, and $5\varphi^{-2q}$. Moreover, $h \geq 3$ and $\lambda^6 > 190$, whence

$$|e^{\Lambda_0} - 1| < 10 \left(\lambda^{-2h} + \varphi^{-2q} \right) < \frac{1}{10}. \quad (120)$$

The inequality $|\log(1+x)| \leq |x|/(1-|x|)$ for $|x| < 1$ now gives the sharper estimate

$$0 < |\Lambda_0| < 12 \left(\lambda^{-2h} + \varphi^{-2q} \right), \quad (121)$$

where Λ_0 is defined in [equation \(86\)](#). Put $w = \min\{h, q\gamma\}$. Since $\log \lambda > 4/5$, [equation \(121\)](#) implies

$$0 < |2q\gamma - n + \mu_\sigma| < 30(\lambda^2)^{-w}. \quad (122)$$

Take

$$M_0 = 2 \cdot 10^{32} \quad (123)$$

and the rational number

$$\frac{p}{Q} = \frac{2585762774943540084566744409566833}{4736007914708481810221186278309584}. \quad (124)$$

Exact rational intervals for square roots and logarithms certify

$$\gcd(p, Q) = 1, \quad Q > 6M_0, \quad 2Q|Q\gamma - p| < 1. \quad (125)$$

Only the certified error is used below. For both signs, the same interval certificate gives

$$\eta_\sigma = \|Q\mu_\sigma\| - M_0|Q\gamma - p| > \frac{7}{100}, \quad (126)$$

and

$$\eta_\sigma(\lambda^2)^{48} > 30Q. \quad (127)$$

Apply [lemma 5.1](#) with

$$a = 2q, \quad b = n, \quad \mu = \mu_\sigma, \quad A = 30, \quad B = \lambda^2, \quad w = \min\{h, q\gamma\}, \quad M = M_0. \quad (128)$$

Equations (125) to (127) then prove $\min\{h, q\gamma\} < 48$. Since $\gamma > 1/2$, either $h < 48$, or $q < 96$ and then

$$h < n < 2q < 192. \quad (129)$$

Thus every target with $q \geq 7$ satisfies

$$3 \leq h < 192, \quad h \text{ odd}. \quad (130)$$

For each of the 190 pairs

$$\sigma \in \{1, -1\}, \quad h \in \{3, 5, \dots, 191\}, \quad (131)$$

put

$$\nu_{\sigma,h} = \frac{\log \mathcal{A}_{\sigma,h}}{\log \lambda}. \quad (132)$$

Equation (102) gives

$$0 < |2q\gamma - n + \nu_{\sigma,h}| < 27(\varphi^2)^{-q}. \quad (133)$$

The same p/Q satisfies, for every one of the 190 shifts,

$$\eta_{\sigma,h} = \|Q\nu_{\sigma,h}\| - M_0|Q\gamma - p| > \frac{9}{1000} \quad (134)$$

and

$$\eta_{\sigma,h}(\varphi^2)^{90} > 27Q. \quad (135)$$

The weakest certified interval occurs at $(\sigma, h) = (1, 31)$ and remains strictly above the displayed margin. Apply lemma 5.1 for each pair with

$$a = 2q, \quad b = n, \quad \mu = \nu_{\sigma,h}, \quad A = 27, \quad B = \varphi^2, \quad w = q, \quad M = M_0. \quad (136)$$

Equations (125), (134) and (135) prove

$$q < 90. \quad (137)$$

The interval proof uses only rational arithmetic. For a positive rational x , logarithms are enclosed using

$$\log x = 2 \sum_{r=0}^{N-1} \frac{z^{2r+1}}{2r+1} + R_N, \quad z = \frac{x-1}{x+1}, \quad (138)$$

with the geometric tail bound

$$|R_N| \leq \frac{2|z|^{2N+1}}{(2N+1)(1-z^2)}. \quad (139)$$

Square roots are bounded between consecutive rational grid points. Hence no binary floating-point comparison enters equations (125) to (127), (134) and (135).

At this point the unbounded even-exponent problem has become finite: every hypothetical target has $q < 90$. What remains is an exact recurrence calculation, not a further asymptotic estimate.

5.4. Exact terminal enumeration. The remaining lower boundary $q = 1$ is direct:

$$D_{1,1} = \lambda, \quad D_{1,-1} = \lambda^{-1}. \quad (140)$$

For the plus sign, the first integral power strictly above $D_{1,1}^2 = \lambda^2$ is $\lambda^3 > 2\lambda^2$. For the minus sign, every positive integral power is at least $\lambda > 2\lambda^{-2} = 2D_{1,-1}^2$. Thus neither sign has a strict candidate at $q = 1$.

It remains to test $2 \leq q < 90$, $3 \nmid q$, and both signs. For each row, exact recurrence arithmetic constructs $D_{q,\sigma}^2$ and the first even Pell unit above it. The algorithm then checks the strict upper window, computes the two coefficient gcd, and tests the exact norm identity. Signs of $a + b\sqrt{2}$ are decided over the integers from a , b , and $a^2 - 2b^2$; no numerical approximation is used.

The complete output consists of

$$q = 2, \sigma = -1 : \quad \lambda^2 - (-F_2 + F_3\sqrt{2})^2 = 6\lambda^{-1}, \quad (141)$$

$$q = 4, \sigma = 1 : \quad \lambda^6 - (F_4 + F_5\sqrt{2})^2 = 40\lambda. \quad (142)$$

No other row passes. In particular, the range $7 \leq q < 90$ is empty; together with [equation \(137\)](#), this proves the even-exponent branch of [theorem 1.1](#). The odd branch cannot use the primitive rectangle or the inequality $h < n$; the next section closes it by reversing the order of the fixed and moving logarithmic-form estimates.

6. THE ODD-EXPONENT BRANCH

The preceding sections classified the even-exponent branch. We now return to the intrinsic setup of [section 2](#) and suppose that the exponent n is odd. The parity selector forces $3 \mid q$, so this branch is disjoint from the primitive-rectangle argument. Its content has a different factorisation, and in particular no analogue of the divisor-height separation [lemma 4.1](#) is available. The proof closes after reversing the order of the two logarithmic-form estimates.

Proposition 6.1 (Negative unit exponent in the odd branch). *Suppose that $q \geq 1$, $\sigma \in \{1, -1\}$, and a positive odd integer n satisfy [equations \(23\)](#) and [\(24\)](#). Then $3 \mid q$, and the unit normalisation has the form*

$$\lambda^n - D_{q,\sigma}^2 = g\lambda^{-h}, \quad h \geq 3 \text{ odd}. \quad (143)$$

Proof. By [proposition 2.2](#), $3 \mid q$ and hence $q \geq 3$. Put

$$A_+ = F_q + F_{q+1}\sqrt{2}, \quad A_- = -F_q + F_{q+1}\sqrt{2}. \quad (144)$$

Both are positive. By [lemma 2.1](#), there is a unique odd integer j such that

$$\lambda^n - D_{q,\sigma}^2 = g\lambda^j. \quad (145)$$

Since $\overline{D_{q,\sigma}} = -D_{q,-\sigma}$ and both n and j are odd, conjugation gives

$$\lambda^{-n} + D_{q,-\sigma}^2 = g\lambda^{-j}. \quad (146)$$

Division of [equation \(145\)](#) by [equation \(146\)](#), followed by the strict upper window, yields

$$\lambda^{2j} < \frac{D_{q,\sigma}^2}{D_{q,-\sigma}^2}. \quad (147)$$

The ratio bounds [lemma 3.1](#) say $\lambda A_- < A_+ < 3A_-$. Thus $j \leq 1$ for $\sigma = 1$, whereas $j \leq -3$ for $\sigma = -1$.

It remains to remove $j = 1$ and $j = -1$ on the plus branch. The following coefficient argument in fact excludes those values for either sign. Set

$$H_+ = F_{q+1}^2 + F_{q+2}^2 = F_{2q+3}, \quad H_- = F_{q+1}^2 + F_{q-1}^2 = 3F_q^2 + 2(-1)^q. \quad (148)$$

If $j = 1$, multiply [equation \(145\)](#) by λ^{-1} and compare the $\sqrt{2}$ -coefficients; if $j = -1$, multiply it by λ instead. The resulting equations are

$$P_{n-1} = H_{-\sigma} \quad \text{or} \quad P_{n+1} = H_{\sigma}, \quad (149)$$

respectively. Alekseyev's complete Fibonacci–Pell intersection theorem states that the common values of the two sequences are 0, 1, 2, 5 [[2](#), Theorem 9]. Hence $H_+ = F_{2q+3} \geq F_9 = 34$ cannot occur. Alekseyev and Tengely's complete near-square table gives

$$3y^2 + 2 \in \{2, 5, 29\}, \quad 3y^2 - 2 \in \{1\} \quad (150)$$

whenever the displayed number is a Pell value [[3](#), Table 1]. If q is odd, then $H_- = 3F_q^2 - 2 \geq 10$; if q is even, then $H_- = 3F_q^2 + 2 \geq 194$. Thus neither alternative in [equation \(149\)](#) is possible. Every remaining j is at most -3 , proving [equation \(143\)](#) with $j = -h$. $\square$

Lemma 6.2 (Odd trace and content factorisation). *Under the hypotheses of [proposition 6.1](#), put*

$$f = F_q, \quad x = F_{q+1}, \quad L = f^2 + 2x^2, \quad R_q = 2x^2 - f^2, \quad M = n + h. \quad (151)$$

Then

$$\lambda^n - \lambda^{-n-2h} = D_{q,\sigma}^2 + D_{q,-\sigma}^2 \lambda^{-2h}, \quad (152)$$

and

$$R_q^2 + 1 = g(2C_M - g), \quad 1 \leq g < R_q, \quad g \mid R_q^2 + 1. \quad (153)$$

In particular, every prime divisor of g is congruent to 1 modulo 4.

Proof. Conjugating [equation \(143\)](#) gives

$$\lambda^{-n} + D_{q,-\sigma}^2 = g\lambda^h. \quad (154)$$

Eliminating g between this equation and [equation \(143\)](#) proves [equation \(152\)](#). Since $\lambda^{-h} = -C_h + P_h\sqrt{2}$, coefficient comparison gives

$$L - C_n = gC_h, \quad P_n - 2\sigma fx = gP_h. \quad (155)$$

Taking norms in [equation \(143\)](#), and using that $M = n + h$ is even, gives

$$R_q^2 + 1 = g(2C_M - g). \quad (156)$$

The Pell addition formulas and [equation \(155\)](#) also give

$$C_M - g = LC_h + 4\sigma fxP_h. \quad (157)$$

This is positive for $\sigma = 1$. For $\sigma = -1$, use

$$L - 3fx = (x - f)(2x - f) > 0, \quad 3C_h > 4P_h \quad (158)$$

to obtain positivity again. Hence $g < C_M$, so $g^2 < R_q^2 + 1$. Here g is odd by the coefficient parity in [equation \(27\)](#), whereas R_q is even because $3 \mid q$. Therefore $g < R_q$, and the divisibility assertion follows from the first factorisation in [equation \(153\)](#). Finally, if a prime p divides g , then p is odd and $R_q^2 \equiv -1 \pmod{p}$; hence $p \equiv 1 \pmod{4}$. $\square$

The factorisation in [lemma 6.2](#) explains why the even proof cannot simply be quoted. It gives no lower bound for g and no inequality $h < n$. We therefore use the fixed logarithmic form first, control $\min\{h, q\gamma\}$, and only then invoke the moving form.

Theorem 6.3 (Odd-exponent exclusion). *For $q \geq 1$ and $\sigma \in \{1, -1\}$, no positive odd integer n can satisfy both [equations \(23\)](#) and [\(24\)](#).*

Proof. Suppose otherwise. By [proposition 6.1](#), $3 \mid q$ and [equation \(143\)](#) holds. We first treat $q \geq 7$; the indices $q = 3, 6$ will be included in the exact terminal enumeration.

Retain φ , $\epsilon = (-1)^q$, and the four Binet coefficients from [equations \(95\)](#) and [\(96\)](#), and put

$$\gamma = \frac{\log \varphi}{\log \lambda}, \quad H = 1 + \log(2q). \quad (159)$$

The window and the elementary bound $D_{q,\sigma} < 4\varphi^q$ give

$$\lambda^n < 32\varphi^{2q} < \lambda^{2q}, \quad n < 2q. \quad (160)$$

Indeed, $\lambda/\varphi > 7/5$ and $(7/5)^{14} > 32$.

We begin with the fixed form

$$\Lambda_0 = n \log \lambda - 2q \log \varphi - \log \left(\frac{u_\sigma^2}{5} \right). \quad (161)$$

Define

$$\delta_\sigma = \epsilon \frac{u'_\sigma}{u_\sigma} \varphi^{-2q}, \quad \tilde{\delta}_\sigma = \epsilon \frac{v'_\sigma}{v_\sigma} \varphi^{-2q}. \quad (162)$$

Expanding [equation \(152\)](#) gives the exact identity

$$\begin{aligned} e^{\Lambda_0} - 1 &= 2\delta_\sigma + \delta_\sigma^2 + \frac{v_\sigma^2}{u_\sigma^2} (1 + \tilde{\delta}_\sigma)^2 \lambda^{-2h} \\ &\quad + \frac{5}{u_\sigma^2} \lambda^{-n-2h} \varphi^{-2q}. \end{aligned} \quad (163)$$

The final sign is positive; this is the precise change from [equation \(118\)](#). The coefficient estimates used there are insensitive to this sign and give

$$0 < |\Lambda_0| < 12 \left(\lambda^{-2h} + \varphi^{-2q} \right). \quad (164)$$

Nonvanishing is exact: if $\Lambda_0 = 0$, full norms in $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ would give $1 = 5^{-4}$. By [lemma 4.2](#) and [equations \(92\)](#) and [\(160\)](#), the same explicit height parameters as in [section 4](#) yield

$$\log |\Lambda_0| > -1.855 \cdot 10^{14} H. \quad (165)$$

Put

$$w = \min\{h, q\gamma\}. \quad (166)$$

The right-hand side of [equation \(164\)](#) is at most $24\lambda^{-2w}$. Since $\log \lambda > 4/5$ and $\log 24 < 4$, comparison with [equation \(165\)](#) gives

$$w < 1.16 \cdot 10^{14} H. \quad (167)$$

We next use the moving coefficient $\mathcal{A}_{\sigma,h}$ from [equation \(97\)](#). The form

$$\Lambda_1 = n \log \lambda - 2q \log \varphi - \log \mathcal{A}_{\sigma,h} \quad (168)$$

has the exact remainder

$$\begin{aligned} \mathcal{R}_{\sigma,q,h}^{\text{odd}} &= \frac{2\epsilon}{5} \left((1 - \sigma\sqrt{2}) + (1 + \sigma\sqrt{2})\lambda^{-2h} \right) \\ &\quad + \frac{\varphi^{-2q}}{5} \left((u'_\sigma)^2 + (v'_\sigma)^2 \lambda^{-2h} \right) + \lambda^{-n-2h}. \end{aligned} \quad (169)$$

Thus

$$\lambda^n = \mathcal{A}_{\sigma,h} \varphi^{2q} + \mathcal{R}_{\sigma,q,h}^{\text{odd}}. \quad (170)$$

Only the sign of the last term differs from [equation \(99\)](#). The same absolute estimates, full norm [equation \(104\)](#), and height bound [equation \(107\)](#) therefore give

$$0 < |\Lambda_1| < 21\varphi^{-2q} \quad (171)$$

and, by [lemma 4.2](#) and [equation \(92\)](#),

$$2q \log \varphi < 9.275 \cdot 10^{12} (4h + 44)H + \log 21. \quad (172)$$

If $q \geq 10^{32}$, then $\gamma > 1/2$ and the monotonicity of q/H show that $q\gamma$ exceeds the right-hand side of [equation \(167\)](#). Hence $w = h$. At $q_0 = 10^{32}$ one has $H(q_0) < 79$. Substitution of $h < 1.16 \cdot 10^{14}H$ into [equation \(172\)](#), using $\log \varphi > 2/5$, gives at q_0

$$q < \frac{5}{4} \left[9.275 \cdot 10^{12} \left(4(1.16 \cdot 10^{14})H + 44 \right) H + 4 \right] < 3.36 \cdot 10^{31}. \quad (173)$$

After division by q , the terms H^2/q , H/q , and $1/q$ decrease for $q \geq q_0$, so the contradiction persists. Consequently

$$q < 10^{32}. \quad (174)$$

It remains to apply the exact reductions already used in [section 5](#). Put

$$\mu_\sigma = \frac{\log(u_\sigma^2/5)}{\log \lambda}. \quad (175)$$

Dividing [equation \(164\)](#) by $\log \lambda$ gives

$$0 < |2q\gamma - n + \mu_\sigma| < 30(\lambda^2)^{-w}, \quad 2q < 2 \cdot 10^{32}. \quad (176)$$

The common rational approximation and the two fixed-shift certificates [equations \(124\)](#) to [\(127\)](#) depend only on γ , μ_σ , and λ , not on the parity of n . The absolute-value reduction in [lemma 5.1](#) therefore proves

$$w < 48. \quad (177)$$

If $q \geq 96$, then $q\gamma > 48$, so $h = w < 48$. Put

$$\nu_{\sigma,h} = \frac{\log \mathcal{A}_{\sigma,h}}{\log \lambda}. \quad (178)$$

[Equation \(171\)](#) gives

$$0 < |2q\gamma - n + \nu_{\sigma,h}| < 27(\varphi^2)^{-q}. \quad (179)$$

The moving certificates [equations \(134\)](#) and [\(135\)](#) cover both signs and every odd $3 \leq h < 192$, hence in particular the present range $3 \leq h < 48$. A second application of [lemma 5.1](#) gives $q < 90$, contradicting $q \geq 96$. We conclude that

$$q < 96. \quad (180)$$

The remaining indices are exactly

$$q = 3, 6, \dots, 93, \quad \sigma \in \{1, -1\}. \quad (181)$$

Because $\lambda > 2$, each strict factor-two window contains at most one integral power. Exact recurrence arithmetic constructs every candidate in [equation \(181\)](#), computes its coefficient content, and tests the norm identity. The complete terminal list is empty. This contradiction proves the theorem. $\square$

Together with the even branch and the direct $q = 1$ check in [equation \(140\)](#), the odd exclusion proves [theorem 1.1](#).

The order of the argument is essential. The fixed Matveev estimate first controls $\min\{h, q\gamma\}$; only in the large- q alternative does this become a bound for h . The moving estimate then bounds q , after which the same fixed and moving rational-interval certificates reduce the problem to the finite domain [equation \(181\)](#). No finite search is extrapolated beyond its proved range.

The all-exponent nearest-gap classification is now complete. We next reinterpret it dynamically, where it becomes the arithmetic input to a complete Pell-orbit theorem.

7. ARITHMETIC DYNAMICS AND COMPLETE PELL ORBITS

The nearest-gap equation is not an isolated algebraic coincidence. It is the exact image of an integral dynamical system under a quadratic-unit semiconjugacy. This section first proves that structural statement for a general seed, then specializes it to the two Fibonacci cores and obtains the complete list in [theorem 1.2](#).

For an integral seed $z = (u, v)$, put

$$\begin{aligned} S(u, v) &= (v, u + 2v), \\ \mathcal{H}(u, v) &= u^2 + v^2, \\ \mathcal{B}(u, v) &= v^2 - u^2 + 2uv, \\ \mathcal{Q}(u, v) &= v^2 - 2uv - u^2, \end{aligned} \quad (182)$$

and define

$$A_z = (v - u) + u\sqrt{2}. \quad (183)$$

Proposition 7.1 (Seed/unit semiconjugacy). *The map $z \mapsto A_z$ is a bijection*

$$\{(u, v) \in \mathbb{Z}^2 : 0 < u \leq v\} \longrightarrow \{a + b\sqrt{2} \in \mathbb{Z}[\sqrt{2}] : a \geq 0, b > 0\}, \quad (184)$$

with inverse $a + b\sqrt{2} \mapsto (b, a + b)$. For every integral seed and every $t \geq 0$,

$$N(A_z) = \mathcal{Q}(z), \quad A_{S^t z} = \lambda^t A_z. \quad (185)$$

Proof. The inverse in [equation \(184\)](#) is immediate. Direct calculation gives

$$N(A_z) = (v - u)^2 - 2u^2 = \mathcal{Q}(z) \quad (186)$$

and

$$A_{S^t z} = (u + v) + v\sqrt{2} = \lambda((v - u) + u\sqrt{2}). \quad (187)$$

Iteration proves [equation \(185\)](#). $\square$

For a positive integral seed, $\mathcal{Q}(z) = 0$ would force $(v - u)^2 = 2u^2$, which is impossible. Thus the canonical case $\mathcal{Q}(z)^2 = 1$ and the noncanonical case $\mathcal{Q}(z)^2 > 1$ exhaust all positive seeds.

In coefficient columns this intertwining is an integral conjugacy. Indeed,

$$M = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} -1 & 1 \\ 1 & 0 \end{pmatrix} \quad (188)$$

are respectively the seed step, multiplication by λ , and the map $(u, v)^T \mapsto (v - u, u)^T$. Direct multiplication gives

$$RM = UR. \quad (189)$$

Since $R \in \mathrm{GL}_2(\mathbb{Z})$, no lattice information is lost: $M^2 \in \mathrm{SL}_2(\mathbb{Z})$ is conjugate to the hyperbolic norm-one unit $\iota(\lambda^2)$.

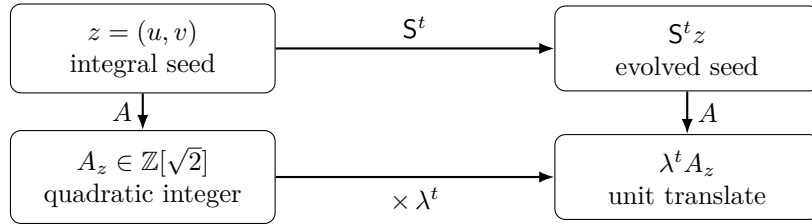


FIGURE 4. The exact seed/unit commutative diagram. The lower dynamics is multiplication in $\mathbb{Z}[\sqrt{2}]$, not an asymptotic model of the upper dynamics. Norm gives the signed covariance $\mathcal{Q}(S^t z) = (-1)^t \mathcal{Q}(z)$; equivalently, $\mathcal{Q}^2$ is invariant.

There is a second, symmetric-square description. Direct expansion gives

$$\mathcal{B}(z)^2 + \mathcal{Q}(z)^2 = 2\mathcal{H}(z)^2, \quad (190)$$

and

$$\begin{pmatrix} \mathcal{H}(S^t z) \\ \mathcal{B}(S^t z) \\ \mathcal{Q}(S^t z) \end{pmatrix} = \begin{pmatrix} C_{2t} & P_{2t} & 0 \\ 2P_{2t} & C_{2t} & 0 \\ 0 & 0 & (-1)^t \end{pmatrix} \begin{pmatrix} \mathcal{H}(z) \\ \mathcal{B}(z) \\ \mathcal{Q}(z) \end{pmatrix}. \quad (191)$$

The integral matrix in [equation \(191\)](#) preserves $2X^2 - Y^2 - Z^2$. Thus the quadratic observables of the seed lie on an integral Lorentz cone, and S acts through an arithmetic subgroup of its orthogonal group. This formulation is useful rather than decorative: the first two coordinates in [equation \(191\)](#) are exactly the proportionality mechanism behind the orbit/gap equivalence below.

Theorem 7.2 (Anchored orbit/gap equivalence). *Let $z = (u, v)$ have positive integral coordinates and $\mathcal{Q}(z)^2 > 1$. Let d be positive and $t \geq 0$, and suppose either $u \leq v$ or $t \geq 1$. Then*

$$\mathcal{H}(S^t z) = P_{d+2t} \quad (192)$$

if and only if there is an integer $g > 0$ such that

$$\lambda^{d-1} - A_z^2 = g\lambda^{-(2t+1)}. \quad (193)$$

At a hit, if

$$D = \mathcal{H}(z) - P_d, \quad E = C_d - \mathcal{B}(z), \quad (194)$$

then

$$D = gP_{2t}, \quad E = gC_{2t}, \quad g = \gcd(D, E). \quad (195)$$

Moreover, [equation \(193\)](#) automatically satisfies

$$A_z^2 < \lambda^{d-1} < 2A_z^2, \quad (196)$$

and its coefficient content and norm are respectively g and $-g^2$.

Proof. Put $w = S^t z$ and $k = d + 2t$. The hypotheses ensure that w is positive and ordered. At a hit, [equation \(190\)](#) and $C_k^2 - 2P_k^2 = (-1)^d$ give

$$\mathcal{B}(w)^2 = C_k^2 - (\mathcal{Q}(z)^2 + (-1)^d) < C_k^2. \quad (197)$$

Because w is positive and ordered, $0 < \mathcal{B}(w) < C_k$. For $t \geq 1$, comparison of the first coordinates in [equation \(191\)](#) with the Pell addition formula gives

$$C_{2t}D = P_{2t}E. \quad (198)$$

The second-coordinate difference is $\mathcal{B}(w) - C_k = -E/C_{2t}$, so $D, E > 0$. Since $\gcd(C_{2t}, P_{2t}) = 1$, this proves [equation \(195\)](#). At $t = 0$, the hit gives $D = 0$ and [equation \(197\)](#) gives $E > 0$; the same formula holds with $P_0 = 0$ and $C_0 = 1$.

The parity-free identity

$$\lambda A_z^2 = \mathcal{B}(z) + \mathcal{H}(z)\sqrt{2} \quad (199)$$

turns [equation \(194\)](#) into

$$\lambda^d - \lambda A_z^2 = E - D\sqrt{2}. \quad (200)$$

At a hit, [equation \(195\)](#) makes the right-hand side $g\lambda^{-2t}$, and division by λ proves [equation \(193\)](#). Conversely, [equations \(193\)](#) and (200) recover [equation \(195\)](#) coefficientwise; the first row of [equation \(191\)](#) then recovers the hit.

It remains to prove the window. Multiplying [equation \(193\)](#) by λ^{2t} and using [equation \(185\)](#) gives

$$\lambda^{k-1} - A_w^2 = g\lambda^{-1}. \quad (201)$$

At the hit, $g = C_k - \mathcal{B}(w)$. Combining this with [equation \(199\)](#) at w gives

$$\lambda A_w^2 - g = 2\mathcal{B}(w) + (-1)^{k+1}\lambda^{-k} > 0. \quad (202)$$

For even k , positivity follows from $2\mathcal{B}(w) \geq 2 > \lambda^{-k}$; for odd k it is immediate. Hence $0 < g\lambda^{-(2t+1)} < A_z^2$, which is [equation \(196\)](#). Finally, $\lambda^{-(2t+1)}$ is a coefficient-primitive norm- -1 unit, proving the content and norm assertions. $\square$

Remark 7.3 (Canonical seeds). If $0 < u \leq v$ and $\mathcal{Q}(z)^2 = 1$, then [proposition 7.1](#) and the unit group of $\mathbb{Z}[\sqrt{2}]$ give a unique $r \geq 1$ with

$$z = (P_r, P_{r+1}), \quad A_z = \lambda^r. \quad (203)$$

Consequently

$$\mathcal{H}(S^t z) = P_{d+2t} \iff d = 2r + 1, \quad (204)$$

and this equality persists for every $t \geq 0$. The corresponding gap is zero, so these seeds are correctly separated from [theorem 7.2](#).

We now specialize to

$$z_{q,\sigma} = (F_{q+1}, F_{q+1} + \sigma F_q). \quad (205)$$

For both signs,

$$A_{z_{q,\sigma}} = D_{q,\sigma}, \quad \mathcal{Q}(z_{q,\sigma}) = F_q^2 - 2F_{q+1}^2. \quad (206)$$

Theorem 7.4 (Complete all-anchor Fibonacci-core orbits). *For every $q \geq 1$, $\sigma \in \{1, -1\}$, positive d , and $t \geq 0$, the hit*

$$\mathcal{H}(S^t z_{q,\sigma}) = P_{d+2t} \quad (207)$$

occurs exactly in the four rows of [equation \(11\)](#).

Proof. First let $q \geq 2$. Then

$$\mathcal{Q}(z_{q,\sigma})^2 - 1 = F_{q-1}F_{q+1}F_{q+2}F_{q+4} > 0. \quad (208)$$

The strict window in [theorem 7.2](#) first forces $d \geq 2$: if $d = 1$, it would give $A_{z_{q,\sigma}}^2 < 1$, whereas $A_{z_{q,\sigma}} > 1$ for $q \geq 2$. Hence an odd anchor here has $d \geq 3$, so $d - 1$ is a positive even exponent.

For the plus sign the seed is ordered, so [theorem 7.2](#) applies at every time. If d is odd, the gap exponent $d - 1$ is even. The only plus target in [theorem 1.1](#) has normalized direction λ , whereas the orbit bridge requires $\lambda^{-(2t+1)}$. Hence there is no plus hit. If d is even, the gap exponent is odd, which is impossible by [theorem 1.1](#). Thus the plus core has no hit for $q \geq 2$.

For the minus sign and $t \geq 1$, the evolved seed is ordered, so the same bridge applies. At an odd anchor, the only minus target has normalized direction λ^{-1} ; it would force $t = 0$, a contradiction. At an even anchor, the exponent is odd and no target exists. Thus the minus core has no positive-time hit.

It remains to classify direct minus hits. The identity

$$\mathcal{H}(z_{q,-1}) = F_{q+1}^2 + F_{q-1}^2 = 3F_q^2 + 2(-1)^q \quad (209)$$

and the complete Pell-value classification used in [proposition 3.2](#) give exactly

$$(q, d) = (2, 3), (4, 5) \quad (210)$$

for $q \geq 2$. These are the last two rows of [equation \(11\)](#).

Finally, $z_{1,1} = (1, 2) = (P_1, P_2)$ is canonical. By [remark 7.3](#), it gives $d = 3$ and every $t \geq 0$. For the minus sign, $z_{1,-1} = (1, 0)$ gives the direct hit $\mathcal{H}(1, 0) = P_1$. If $t \geq 1$, then

$$S^t(1, 0) = (P_{t-1}, P_t), \quad \mathcal{H}(S^t(1, 0)) = P_{2t-1} < P_{d+2t}, \quad (211)$$

so there are no further hits. This proves the exhaustive list. $\square$

Remark 7.5 (Exact scope). The theorem classifies both signs, every positive anchor, and every nonnegative time for the displayed Fibonacci cores. It does not classify negative time or all noncanonical positive seeds with $\mathcal{Q}(z)^2 > 1$. Most importantly, no result here shows that a hypothetical Markoff collision must produce one of these cores. The semiconjugacy closes the downstream arithmetic problem; it does not supply the missing upstream Markoff reduction.

The next section changes perspective. Instead of classifying the global equation, it asks how much of its divisibility can recur locally, and thereby identifies the precise limits of p -adic information.

8. LOCAL CLOCKS, HAAR RECURRENCE, AND THEIR LIMITS

The global classification uses the full norm–content equation. A tempting alternative is to control only the common prime–power content forced by the Fibonacci and Pell recurrences. We now show exactly how far that local idea can go. Compatible p -adic clocks recur on positive-density subfamilies of genuine nearest windows, but a rank-parity obstruction can also prevent a split prime from occurring at all.

For the plus sign put

$$D_q = F_q + F_{q+1}\sqrt{2}, \quad m(q) = \lceil \log_\lambda D_q \rceil. \quad (212)$$

Whenever $3 \nmid q$ and

$$D_q < \lambda^{m(q)} < \sqrt{2}D_q, \quad (213)$$

the even power $\lambda^{2m(q)}$ lies in the strict nearest window for D_q^2 . Put $s = (-1)^{m(q)}$ and define

$$G(q, m) = \gcd(R_q - s, 2F_{q+1}P_m - F_qC_m). \quad (214)$$

This cross-gcd is defined on every row in [equation \(213\)](#); the norm–content equation is not assumed.

Lemma 8.1 (Two exact phase factorizations). *For $3 \nmid q$,*

$$\begin{aligned} m \equiv q \pmod{2} &\implies G(q, m) = \gcd(F_{q+1}, C_m) \gcd(F_{q+2}, C_{m+1}) \\ m \not\equiv q \pmod{2} &\implies G(q, m) = \gcd(F_{q-1}, C_{m-1}) \gcd(F_{q+4}, C_{m+2}). \end{aligned} \quad (215)$$

Proof. Put $\varepsilon = (-1)^q$ and $E = 2F_{q+1}P_m - F_qC_m$. Cassini’s identity gives

$$\begin{aligned} R_q - \varepsilon &= F_{q+1}F_{q+2}, \\ R_q + \varepsilon &= F_{q-1}F_{q+4}. \end{aligned} \quad (216)$$

Direct reduction gives

$$\begin{aligned} E &\equiv -F_qC_m \pmod{F_{q+1}}, \\ E &\equiv -F_qC_{m+1} \pmod{F_{q+2}}, \\ E &\equiv F_qC_{m-1} \pmod{F_{q-1}}, \\ 3E &\equiv -F_qC_{m+2} \pmod{F_{q+4}}. \end{aligned} \quad (217)$$

The equal-phase Fibonacci factors are coprime. In the unequal phase,

$$\gcd(F_{q-1}, F_{q+4}) \in \{1, 5\},$$

but the exceptional prime 5 divides no companion number C_j : the recurrence modulo 5 has period 12 and nonzero residues 1, 1, 3, 2, 2, 1, 4, 4, 2, 3, 3, 4. The only possible failure of the displayed multipliers to be units is at 3. Here

$$3 \mid F_r \iff 4 \mid r, \quad 3 \mid C_j \iff j \equiv 2 \pmod{4},$$

and every aligned Fibonacci–companion index pair has opposite parity. Thus 3 cannot occur in an aligned gcd. The product identity follows prime by prime and valuation by valuation. $\square$

The factorisation replaces one two-coordinate gcd by two aligned recurrence clocks. We now determine exactly which primes can synchronize in one of these clocks to arbitrary depth.

Let p be an odd prime with $(2/p) = 1$ that divides some C_j . Define

$$\begin{aligned} z_F(p) &= \min\{r > 0 : p \mid F_r\}, & e_F(p) &= v_p(F_{z_F(p)}), \\ z_C(p) &= \min\{r > 0 : p \mid C_r\}, & e_C(p) &= v_p(C_{z_C(p)}), \end{aligned} \quad (218)$$

and, for $a \geq 1$,

$$B_p(a) = z_F(p)p^{\max(0, a - e_F(p))}, \quad J_p(a) = z_C(p)p^{\max(0, a - e_C(p))}. \quad (219)$$

The exact rank laws are

$$p^a \mid F_r \iff B_p(a) \mid r, \quad p^a \mid C_j \iff j \equiv J_p(a) \pmod{2J_p(a)}. \quad (220)$$

The first is the standard Fibonacci rank and valuation law [12, 17]. For the second, choose a Hensel lift $\sqrt{2} \in \mathbb{Z}_p$, put $u = 1 + \sqrt{2}$ and $b = -u^2$, and use

$$2u^j C_j = (-1)^j (b^j + 1). \quad (221)$$

Companion zeros are the odd multiples of $z_C(p)$, and odd-prime lifting gives their complete valuations. In local-field language, $\mathbb{Q}(\sqrt{2}) \otimes \mathbb{Q}_p \simeq \mathbb{Q}_p \times \mathbb{Q}_p$ at such a split prime, and [equation \(221\)](#) converts the clock into an order and valuation statement for a unit of $\mathbb{Z}_p$; this is the special case of the standard framework described by Serre [18].

Proposition 8.2 (Local-to-Archimedean transfer). *Let $\mathcal{P}$ be a finite nonempty set of primes satisfying the preceding hypotheses, prescribe $a_p \geq 1$, and partition $\mathcal{P} = \mathcal{P}_1 \sqcup \mathcal{P}_2$. Choose $\delta \in \{0, 1\}$ and the offsets*

δ	(r_1, t_1)	(r_2, t_2)
0	(1, 0)	(2, 1)
1	(-1, -1)	(4, 2)

(222)

Suppose that integers q_0, m_0 satisfy

$$3 \nmid q_0, \quad q_0 - m_0 \equiv \delta \pmod{2}, \quad (223)$$

and, for $i \in \{1, 2\}$ and $p \in \mathcal{P}_i$,

$$q_0 + r_i \equiv 0 \pmod{B_p(a_p)}, \quad m_0 + t_i \equiv J_p(a_p) \pmod{2J_p(a_p)}. \quad (224)$$

Put

$$Q = \text{lcm}(6, \{B_p(a_p) : p \in \mathcal{P}\}), \quad M = \text{lcm}(\{2J_p(a_p) : p \in \mathcal{P}\}). \quad (225)$$

Then infinitely many $q = q_0 + Q\ell$ satisfy [equation \(213\)](#), $m(q) \equiv m_0 \pmod{M}$, and

$$\prod_{p \in \mathcal{P}} p^{a_p} \mid G(q, m(q)), \quad (226)$$

with every prime power carried by its prescribed factor in [equation \(215\)](#). Among the progression $q = q_0 + Q\ell$, this selected family has relative natural density

$$\frac{\vartheta}{M}, \quad \vartheta = \log_\lambda \sqrt{2}. \quad (227)$$

Its counting function up to $q \leq X$ is therefore $\vartheta X/(MQ) + o(X)$.

Proof. The congruences and [equation \(220\)](#) place each prescribed prime power in the required Fibonacci–companion pair of [equation \(215\)](#). They remain valid on the progression $q = q_0 + Q\ell$.

It remains to meet the Archimedean window with the required value of m . Put

$$\alpha = \frac{\log \varphi}{\log \lambda}, \quad \beta = \log_\lambda \left(\frac{1 + \sqrt{2}\varphi}{\sqrt{5}} \right). \quad (228)$$

Binet’s formula gives

$$\log_\lambda D_q = q\alpha + \beta + O(\varphi^{-2q}). \quad (229)$$

The irrationality of α was proved in [theorem 2.3](#); hence $Q\alpha/M$ is irrational. Weyl equidistribution on the circle $\mathbb{R}/M\mathbb{Z}$ shows that the limiting phase lies in the cyclic interval $(m_0 - \vartheta, m_0)$ with normalized Haar measure ϑ/M . Its boundary has Haar measure zero, and the Binet error tends to zero. Membership in this interval is exactly $m(q) \equiv m_0 \pmod{M}$ together with [equation \(213\)](#). This proves the divisibility and both density assertions. $\square$

Theorem 8.3 (Exact clock criterion and simultaneous products). *Let p be an odd prime with $(2/p) = 1$ that divides some C_j . Then the following are equivalent:*

- (i) $z_F(p)$ and $z_C(p)$ are not both even;
- (ii) for every $a \geq 1$, infinitely many nearest-window rows have $p^a \mid G(q, m(q))$;
- (iii) at least one such row has $p \mid G(q, m(q))$.

Every prime $p \equiv 7 \pmod{8}$ satisfies these conditions. Consequently, for every finite set $\mathcal{P}$ of such primes and arbitrary exponents $a_p \geq 1$, infinitely many rows satisfy the one-bin divisibility

$$\prod_{p \in \mathcal{P}} p^{a_p} \mid \gcd(F_{q+1}, C_{m(q)}) \mid G(q, m(q)). \quad (230)$$

In particular, for every $a, b \geq 1$, infinitely many rows satisfy

$$7^a 17^b \mid G(q, m(q)). \quad (231)$$

The rank-parity condition is sharp: $z_F(241) = 120$ and $z_C(241) = 20$, so

$$241 \nmid G(q, m) \quad (232)$$

for every plus nearest-window row.

Proof. For a single prime, use the first factor in the appropriate phase and write $q = B_p(a)u - 1$, with $m \equiv J_p(a) \pmod{2J_p(a)}$. The phase condition becomes

$$B_p(a)u \equiv J_p(a) + 1 \pmod{2}. \quad (233)$$

If $B_p(a)$ is even, this is soluble exactly when $J_p(a)$ is odd; if $B_p(a)$ is odd, the parity of u is free. Multiplication by powers of the odd prime does not change the rank parities, so the congruences are consistent exactly when $z_F(p)$ and $z_C(p)$ are not both even. Choose the residue of u modulo 3 so that $3 \nmid B_p(a)u - 1$. If $3 \mid B_p(a)$ this is automatic; otherwise exactly one residue is forbidden. This choice is compatible with the required parity by the Chinese remainder theorem modulo 6. [Proposition 8.2](#) then gives infinitely many rows at every depth. Conversely, the Fibonacci and companion indices paired in [equation \(215\)](#) have opposite parity. If both ranks are even, no pair can contain p . This proves the equivalence.

Now let $p \equiv 7 \pmod{8}$ and choose $r^2 = 2 \pmod{p}$. For $u = 1 + r$, the order of u has 2-part at most 2. If h is its odd part, then $u^h = \pm 1$; since $1 - r = -u^{-1}$, $C_h \equiv 0 \pmod{p}$. Thus $z_C(p)$ is odd, and the criterion applies. The assertion for $\mathcal{P} = \emptyset$ follows from [theorem 2.3](#), so suppose henceforth that $\mathcal{P} \neq \emptyset$. For finitely many such primes, the lifted companion ranks remain odd. Put

$$B = \text{lcm}_{p \in \mathcal{P}} B_p(a_p), \quad J = \text{lcm}_{p \in \mathcal{P}} J_p(a_p).$$

Then $m \equiv J \pmod{2J}$ satisfies every odd companion class, and $q = Bu - 1$ satisfies every first-bin Fibonacci class. Choose the parity of u to make $q \equiv m \pmod{2}$, and choose its residue modulo 3 as above. The choices are compatible modulo 6, so [proposition 8.2](#) proves [equation \(230\)](#). There are infinitely many primes $7 \pmod{8}$: if $p_1, \dots, p_k$ were all of them, an even multiple A of their product would make $2A^2 - 1 \equiv 7 \pmod{8}$; every prime divisor r has $(2/r) = 1$, hence $r \equiv 1$ or $7 \pmod{8}$, and at least one must be $7 \pmod{8}$. It is new because $2A^2 - 1 \equiv -1 \pmod{p_i}$ for every i .

For [equation \(231\)](#), the exact lifted ranks are

$$\begin{aligned} B_7(a) &= 8 \cdot 7^{a-1}, & J_7(a) &= 3 \cdot 7^{a-1}, \\ B_{17}(b) &= 9 \cdot 17^{b-1}, & J_{17}(b) &= 4 \cdot 17^{b-1}. \end{aligned} \tag{234}$$

Assign 7 and 17 to the two equal-phase bins. Their q moduli are coprime and their m moduli have gcd 2 with odd prescribed residues, so the systems are compatible. More explicitly, $q \equiv -1 \pmod{B_7(a)}$ makes q odd, both companion congruences make m odd, and $q \equiv -2 \pmod{B_{17}(b)}$ gives $q \equiv 1 \pmod{3}$ because $9 \mid B_{17}(b)$. Thus the phase and $3 \nmid q$ hypotheses of [proposition 8.2](#) hold. Finally, the exact recurrence certificates at 241 give the two even ranks displayed in the theorem; the already proved criterion yields [equation \(232\)](#). $\square$

Remark 8.4 (What the clocks establish). The rows above satisfy genuine strict nearest windows, and the density is a Haar-measure statement on a phase circle. They are not asserted to satisfy [equation \(24\)](#). Thus local abundance produces neither a target nor a Markoff counterexample. Its exact conclusion is methodological: uniform boundedness, squarefreeness, finitely many compatible rank clocks, and the assertion that every split prime occurs are all false as global substitutes for the norm-content equation.

This local-global contrast is the organising principle of the final discussion, which situates the exact theorems relative to the parent Markoff problem.

9. SCOPE, CONNECTIONS, AND THE NEXT OBSTRUCTION

The first arithmetic conclusion of this paper is complete on its stated domain: for every $q \geq 1$, both signs, and every positive exponent, the strict Fibonacci–Pell norm-content system has exactly the two rows in [equation \(7\)](#). The proof is unbounded in q and closes only after the analytic bounds have reduced both parity branches to exhaustive finite

domains. The second conclusion is likewise complete: for both signed Fibonacci cores, every positive Pell anchor, and every nonnegative time, [theorem 7.4](#) gives all orbit hits.

The surrounding results explain why these classifications are rigid. At the Archimedean place, candidate windows are abundant and their exact densities are Haar measures of interval intersections on a phase circle. At split finite places, compatible Fibonacci and companion ranks can synchronize to arbitrary prescribed depth. Neither phenomenon sees the oriented primitive unit direction. The global norm–content equation sees both coefficient coordinates simultaneously, and it is this global orientation that enables the primitive rectangle and the logarithmic-form argument. The memorable separation is therefore not local versus computational, but local and measure-theoretic abundance versus global Diophantine rigidity.

The algebraic structures exposed here have precise roles. The norm torus, its negative-norm torsor, the integral unit group, and the Lorentzian symmetric-square representation all act directly in the proofs. The biquadratic field $\mathbb{Q}(\sqrt{2}, \sqrt{5})$ supplies the two conjugations used in the norm and height arguments. These are the cross-field structures needed by the present proof. Further frameworks would become relevant only after an explicit map or invariant connects them to the classified system.

There is, however, a genuine algebraic-geometric object behind the parent problem. Over $\mathbb{Z}$, the Markoff equation defines the affine cubic scheme

$$\mathcal{M} = \operatorname{Spec} \mathbb{Z}[a, b, c] / (a^2 + b^2 + c^2 - 3abc), \quad (235)$$

and the Vieta involutions, such as

$$(a, b, c) \longmapsto (a, b, 3ab - c), \quad (236)$$

are integral automorphisms of this cubic. After the scaling $(x, y, z) = (3a, 3b, 3c)$, its equation becomes $x^2 + y^2 + z^2 - xyz = 0$, the Fricke trace surface with commutator trace -2 . This description is also a concrete algebraic-topological bridge. The fundamental group of a one-holed torus is the free group F_2 ; the three coordinates are traces of two generators and their product in an SL_2 -representation. Goldman shows that the reflection

$$(x, y, z) \longmapsto (x, y, xy - z)$$

comes from the extended mapping-class group of the surface [9, Sections 2.1–2.2 and Appendix A-3]. Scaling back gives exactly [equation \(236\)](#). On the discrete side, regarding positive Markoff triples as vertices and Vieta moves as edges gives the Markoff mutation graph and its familiar descent tree; Christoffel words and integral necklaces give combinatorial encodings of its arithmetic vertices [16, 8]. Thus topology explains the symmetry, graph theory records its orbits, and combinatorics labels the vertices. These are exact structural connections, but not hidden inputs: this paper constructs no morphism from a hypothetical collision locus on $\mathcal{M}$ to the Fibonacci nearest-gap system.

The most immediate arithmetic extension starts instead from [theorem 7.2](#). For every noncanonical positive seed, an orbit hit is equivalent to a strict norm–content gap for the seed-dependent quadratic integer A_z . The Fibonacci cores form one coefficient locus on which the present paper achieves a complete classification. Noncanonical cores with $Q(z)^2 > 1$ lead to more general continuant-dependent coefficients. The next problem is to reconstruct enough primitive geometry from those coefficients to retain effective heights without assuming the very collision one hopes to exclude. That problem is suitable for a separate paper because it requires new arithmetic, not additional exposition around the two-row theorem.

The moving-coefficient logarithmic form should also be understood at the level of mechanism. Grouping all terms of the same dominant growth before applying a logarithmic-form estimate is established practice in exponential Diophantine equations [6, 4]. What is specific here is the coefficient [equation \(97\)](#), its exact full norm [equation \(104\)](#), and the certified reduction that turns those identities into a complete classification. We make no priority claim for the general strategy.

The Markoff-facing bottleneck is therefore sharp. Downstream of the Fibonacci-core hypothesis, the gaps and all positive-anchor orbits are classified. Upstream, one must prove that a repeated largest Markoff number yields one of these cores, find a competing core, or classify the seed-dependent gaps with $Q(z)^2 > 1$.

The frontier is precise: local abundance versus global rigidity. Progress on Markoff uniqueness requires crossing the missing upstream arrow in [figure 1](#).

APPENDIX A. CERTIFICATE ARCHITECTURE AND FINITE COVERAGE

This appendix specifies exactly where computation enters the proof and how the finite conclusions cover the original unbounded statement. The corresponding source files and one-command driver are listed in the supplement.

TABLE 1. Computer-assisted inputs and controls. Finite density and orbit regressions test identities and boundaries but do not prove the corresponding unbounded theorems.

Claim	Complete input	Independent control
Leading exponents	Published complete Fibonacci–Pell and Pell near-square classifications	Two optional Magma V2.29-9 quartic runs and exact arithmetic filters
Even classification	Exact logarithmic-form identities, outward rational intervals, all 190 moving shifts, and enumeration of $2 \leq q < 90$	Two separately structured fixed-point implementations and a wider exact regression
Odd exclusion	Exact changed-sign trace identities, unchanged fixed and moving shifts, and enumeration of $q = 3, 6, \dots, 93$	A source checker and a non-importing integer-arithmetic reconstruction
Window densities	Irrational-rotation proof and exact interval lengths	Exact finite counts through $q = 5000$, used only as regression evidence
All-anchor orbits	Exact semiconjugacy, anchored equivalence, all-exponent theorem, and published direct-hit classifications	Separate general-seed, odd-anchor, and even-anchor identity regressions
Uniform local clocks	Written p -adic rank, Chinese remainder, and equidistribution proof	Exact rank certificates, obstruction at 241, and displayed witnesses

A.1. Logical dependency. No finite computation proves an unbounded statement by extrapolation. In the even branch, the analytic argument applies to every hypothetical target with $q \geq 7$: [propositions 3.2](#) and [4.3](#) and the moving-coefficient form first give $q < 10^{32}$; the certified reductions then give $h < 192$ and $q < 90$. Only after those deductions does the terminal enumeration inspect the complete finite domain

$$2 \leq q < 90, \quad 3 \nmid q, \quad \sigma \in \{1, -1\}. \quad (237)$$

The omitted lower boundary is analytic rather than computational: [equation \(140\)](#) proves directly that neither sign has a strict candidate at $q = 1$. Thus $q = 90$ is excluded by the strict analytic bound, while every lower index in this parity branch and both signs are covered. In the odd branch, the fixed-form-first argument gives $q < 10^{32}$, the fixed reduction gives $\min\{h, q\gamma\} < 48$, and the moving reduction excludes $q \geq 96$. It leaves exactly

$$q = 3, 6, \dots, 93, \quad \sigma \in \{1, -1\}. \quad (238)$$

Exact recurrence arithmetic checks both complete domains. The even output consists only of equations (141) and (142), while the odd output is empty. This proves both existence and completeness of theorem 1.1.

A.2. Rational interval model. All reduction comparisons are made with rational endpoints. Positive square roots are bracketed by integer squaring. After range reduction, logarithms are evaluated through equations (138) and (139). A certificate is accepted only when the lower endpoint of each η is positive and the lower endpoint of the left side in equations (127) and (135) exceeds the upper endpoint of its right side. Strict inequalities therefore remain strict under rounding.

The common integers p, Q are printed in equation (124). The certificate independently verifies $\gcd(p, Q) = 1$ and the approximation error; trust in a floating-point continued-fraction library is not part of the proof. The moving stage evaluates every one of the 190 pairs, rather than sampling or replacing them by a numerical minimum.

A.3. Exact arithmetic model. An element of $\mathbb{Z}[\sqrt{2}]$ is stored as a pair of arbitrary-precision integers (a, b) representing $a + b\sqrt{2}$. Addition, multiplication, conjugation, norm, and coefficient content are consequently exact. The sign of $a + b\sqrt{2}$ is decided from integer signs and the comparison of a^2 with $2b^2$. Fibonacci and Pell coordinates are generated by their integer recurrences. The terminal search therefore contains no floating-point predicate.

The two all-exponent programs share the published classification inputs and the fixed rational-interval constants already displayed in the manuscript. The second independently reconstructs the parity selector, the changed odd trace, the bootstrap arithmetic, and the terminal integer enumeration; it does not import the first. Wider runs through $q = 2000$ are regressions only.

A.4. Published near-square classification. The proof of equation (62) imports the published complete classification in [3, Table 1]. That paper reduces the relevant Lucas near-square equations through complete computational reductions via elliptic curves or Thue equations. Because its table does not publish row-specific transcripts, the supplement retains a separate complete calculation of the two induced quartics. The local Magma transcripts report the complete integral-point routine and the required Mordell–Weil information; standard-library programs then verify every returned point and the exact filtering to equation (62). The manuscript relies on the published theorem for completeness and treats the local calculation as independent corroboration.

A.5. Reproduction. From the manuscript directory, run

```
python3 supplement/run_all.py
```

This executes every standard-library certificate used by the paper. Magma is optional: its inputs and recorded transcripts can be inspected without the proprietary executable, while a fresh Magma V2.29-9 installation can rerun them. The supplement README records the software assumptions, source-to-claim map, expected summaries, and the distinction between proof certificates and non-probative regression checks.

CODE AVAILABILITY

The LaTeX sources, supplementary proof certificates, recorded reference outputs, and reproduction instructions accompanying this article are publicly available in the companion GitHub repository:

<https://github.com/dutykh/fibonacci-pell-nearest-gaps/>

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MATHEMATICS DEPARTMENT, KHALIFA UNIVERSITY OF SCIENCE AND TECHNOLOGY, PO BOX 127788, ABU DHABI, UNITED ARAB EMIRATES
Email address: `denys.dutykh@ku.ac.ae`

UNIV. SAVOIE MONT BLANC, CNRS, LAMA, 73000 CHAMBÉRY, FRANCE
Email address: `laurent.vuillon@univ-smb.fr`